

Reflected Brownian Motion with Drift in a Wedge

DISSERTATION

Submitted in Partial Fulfillment of
the Requirements for
the Degree of

DOCTOR OF PHILOSOPHY

at the

NEW YORK UNIVERSITY
LEONARD N. STERN SCHOOL OF BUSINESS

by

Ziran Liu

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Ziran Liu

September 2023

To my family

ABSTRACT

Reflected Brownian Motion with Drift in a Wedge

by

Ziran Liu

Advisors: Prof. Josh Reed, Ph.D. and Prof. Peter Lakner, Ph.D.

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An intriguing but classical question in probability theory and Operations Research (OR) is the “heavy-traffic” limit of some particularly featured queuing processes, also known as the diffusion approximation in literature. It can provide insight into (mostly the global) behavior of a queueing process, by studying its (functional) limit in a “right” sense (i.e., there can be multiple ways of taking limit in a suitable probabilistic or even deterministic sense). In particular, always the value of the pre-limit queueing processes has some special properties such as non-negativeness, which leads to the most prominent feature of its limiting process of being “constrained” in a domain instead of free in the whole space. A classical example of such a phenomenon is the limiting processes being constrained in the first orthant in the Euclidean space. Therefore the limiting process has to be “reflected” at the boundary of a domain, thus we call this reflected process. (The word “reflect” is used for historical

reasons, although it only means the process is constrained in the domain and blocked by the boundary from going out, instead of the usual meaning of “reflection” in physics.)

Contribution: This PhD research investigates such kind of limiting processes but in a more general and non-traditional setting, i.e., with the state space of a wedge, and with oblique reflection, compared to the traditional case of the state space being the first orthant. The problem under investigation is formulated as the submartingale problem with drift term in a wedge, which is a way of the mathematical formulation of the reflected Brownian motion with drift. This is also a generalization of the work by Varadhan and Williams in 1985. But a key difference in this Ph.D. research is to include the non-zero drift term. Methodologically, the drift makes the explicit solution to a concerned Laplace equation (a key ingredient in the method in Varadhan and Williams in 1985) not exist, nonetheless it makes the solution’s analytic property become unclear and varies from case to case. Physically, the drift can essentially change the long-time behavior of the reflected process, which can beneficially furthermore make the stationary distribution exist. Here we prove the existence and uniqueness properties of this submartingale problem with drift term, strong Markov property, and two types of different Feller properties which are stronger than that in Varadhan and Williams in 1985 even when drift is zero. As well, we provide quantitative properties such as the hitting probability of the vertex of the wedge for the absorption processes and the connection between this solution and the Skorokhod problem.

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Chapter 1

Background knowledge and motivations

1.1 Queueing theory

Queueing Theory is a field in probability theory. It has been widely studied, both in the theoretical side of itself and in its applications. For its nature, it always shows up in a universal collection of scenarios, such as economic models, tele-communication networks, industrial manufacture, computer systems, machine plants, etc.

In this section, we give an introduction to queueing theory. We will present the main definitions and results according to our purpose to further introduce heavy-traffic limit theorems and then reflected Brownian motions (RBM).

We begin by the basic objects in the study of Queueing Theory.

Let's take the classic customer-server model for convenience of explanation.

The queue-length (i.e., the number of customers in the queue) will be in the center of the concern in our study of queueing theory. Apparently, the queue-length is affected by two factors: the **arrival** of a customer into the queue and the **departure** of a customer from the queue.

The arrival of the sequence of customers is random, in queueing theory it is always characterized by characterizing the inter-arrival time or the number of customers arrived in a fixed amount of time. The idea is simple, one is to fix the number of arrival of customers and then we study the distribution of the inter-arrival time (the time needed for ONE new customer to arrive), the other is to fix the amount of time and see how many customer can arrive and characterize by giving its distribution.

To move on, we need the following **Kendall's notation**, which denotes the characteristics of a queue (or a queueing process).

Kendall's notation is a standard system used to describe and classify queueing models. This notation provides essential information about the primary characteristics of a queue.

In general, a queueing model in Kendall's notation is represented as:

$$A/S/c/K/m/Z$$

, in which the last three terms are optional, so we usually use it in a form of

$$A/S/c$$

.

Here's a breakdown of each component:

1. **A - Arrival Process:** Describes the statistical nature of the arrival process.
 - *M*: Markovian or memoryless process (typically modeled as a Poisson process).
 - *D*: Deterministic arrivals (constant time between successive arrivals).
 - *G*: General distribution (no specific distribution is assumed).
2. **S - Service Time Distribution:** Describes the statistical distribution of the service times.

- M : Memoryless service times (exponentially distributed).
 - D : Deterministic service times.
 - G : General distribution.
3. c - **Number of Servers**: A number representing how many servers/channels are in the system.
 4. K - **System Capacity (optional)**: Specifies the total number of entities (in service and in the queue) that the system can handle. If this component is omitted, it's assumed the system has infinite capacity.
 5. m - **Population Size (optional)**: Represents the source population size, or the total number of potential entities that could enter the system. If omitted, it's assumed the source population is infinite.
 6. Z - **Queue Discipline (optional)**: Describes the order or discipline in which entities in the queue are served.
 - $FIFO$ or $FCFS$: First In First Out (or First Come First Serve) — standard queue behavior.
 - $LIFO$: Last In First Out.
 - $SIRO$: Service in Random Order.
 - PRI : Priority service.

In practice, the most common notation to come across is the $A/S/c$ format. The other parameters (like system capacity, population size, and queue discipline) are often omitted unless necessary for the context.

Some commonly seen examples include:

- $M/M/c$: A c servers queue where both the arrival process and service times are Markovian. This is one of the most basic and widely studied queueing models.

- $M/D/c$: A c servers queue where arrivals are Markovian, but service times are deterministic.
- $M/G/c$: A c servers queue where arrivals are Markovian, but service times follow a general distribution.

Now we give some details of concepts and basic results of queueing theory for $M/M/1$ as a toy model .

1.2 M/M/1 queue

An $M/M/1$ queue is a model characterized by **single-server**, **Poisson arrival process**, and **exponential service times**.

Arrival Process: Let $N(t)$ be the number of arrivals in the interval $[0, t]$. The inter-arrival times are assumed to be exponentially distributed with parameter λ . Thus, $N(t)$ is a Poisson process with rate λ .

Service Times: Let S_i be the service time of the i^{th} customer. We assume S_i to be exponentially distributed with parameter μ , and independent of $N(t)$ and other S_j .

Number of Servers: There is a single server.

1.2.1 Key Parameters and Performance Measures

λ : Average rate of customer arrivals.

μ : Average rate at which a single server provides service.

$\rho = \frac{\lambda}{\mu}$: Server utilization.

Performance measures of interest:

L : Expected number of customers in the system.

L_q : Expected number of customers in the queue.

W : Expected waiting time in the system for a customer.

W_q : Expected waiting time in the queue for a customer.

1.2.2 Equilibrium Equations

To derive the steady-state probabilities, consider the balance equations. Let p_n be the probability of having n customers in the system in the steady state.

Using the flow balance principle:

$$\lambda p_{n-1} = \mu p_n, \quad \text{for } n \geq 1 \quad (1.1)$$

From the above, we get:

$$p_n = \left(\frac{\lambda}{\mu}\right)^n p_0 = \rho^n p_0, \quad \text{for } n \geq 1 \quad (1.2)$$

Where p_0 is the probability of having no customers in the system, and can be found using the normalization condition:

$$p_0 + p_1 + p_2 + \cdots = 1 \quad (1.3)$$

1.2.3 Furthermore on Performance Measures

Using the probabilities p_n :

$$L = \sum_{n=0}^{\infty} n p_n = \rho p_0 + 2\rho^2 p_0 + \cdots = \frac{\rho}{1 - \rho} \quad (1.4)$$

From Little's formula:

$$L = \lambda W \implies W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda} \quad (1.5)$$

Similarly, for the queue:

$$L_q = L - \rho = \frac{\rho^2}{1 - \rho} \quad (1.6)$$

$$W_q = \frac{L_q}{\lambda} = \frac{\rho}{\mu - \lambda} \quad (1.7)$$

1.3 Heavy-traffic conditions

In the study of queueing systems, understanding the system's behavior when it operates near its capacity is of great importance. Such scenarios, often termed as the 'heavy traffic' regime, provide insights into the potential bottlenecks, system resilience, and performance degradation as workloads approach system limits. As modern systems, from communication networks to manufacturing pipelines, increasingly push their operational boundaries, the heavy traffic analysis becomes a critical tool. This paper delves deep into the realm of heavy traffic in queueing processes, shedding light on the intricacies of system performance in near-capacity conditions, and offering analytical approximations that serve as vital instruments for system design, optimization, and management.

1.3.1 Heavy Traffic in Queueing Processes

Definition 1. *A queueing system is said to be in a heavy traffic regime when its workload is approaching the system's capacity. Formally, let λ be the arrival rate of customers to the*

system and μ be the service rate. The traffic intensity, denoted by ρ , is defined as:

$$\rho = \frac{\lambda}{\mu}$$

A queueing system is in a heavy traffic condition when ρ approaches 1, i.e. $\rho \approx 1$, implying that the system's capacity is nearly saturated.

The **significance** of heavy-traffic: Studying queueing processes in heavy traffic is crucial because it allows us to understand the system's behavior under near-capacity conditions. Under heavy traffic, the variability in the queueing process and the associated waiting times often become pronounced, leading to potential service degradation.

The **Objective of Study** of heavy-traffic queueing processes: The primary aim of studying queueing systems in heavy traffic is to derive diffusion approximations. These approximations involve characterizing the behavior of the queue length or waiting time process as a stochastic differential equation, typically driven by Brownian motion or some related stochastic process.

The common frameworks are:

1. **Functional Central Limit Theorems (FCLT):** Under heavy traffic conditions, appropriately centered and scaled queue lengths or waiting times can be shown to converge weakly to a specific stochastic process. This convergence is typically established using functional central limit theorems.
2. **Diffusion Approximations:** Given the weak convergence from FCLT, the limiting stochastic process (usually a form of reflected Brownian motion) serves as a diffusion approximation for the queueing process under heavy traffic. Such approximations facilitate the computation of performance metrics like queue lengths, waiting times, and blocking probabilities in near-capacity regimes.

The importance of the study of the heavy-traffic status is also in its **applications**: the insights derived from heavy traffic studies enable better system design, capacity planning, and dynamic resource allocation. By understanding system behavior in its limits, we can design mechanisms to handle peak loads, ensuring system resilience and efficient utilization.

1.4 Diffusion approximations

We have already known the heavy traffic conditions, now we give two examples as to show the “limit theorems” under this kind of heavy traffic conditions.

1.4.1 $M/M/1$ Queue

Theorem 1.4.1. *Consider an $M/M/1$ queueing system with arrival rate λ and service rate μ . Let $Q(t)$ be the number of customers in the system at time t , and define the traffic intensity $\rho = \frac{\lambda}{\mu}$. If we denote $B(t)$ as a standard Brownian motion, then as the traffic intensity $\rho \rightarrow 1$, the scaled and centered queue length process converges in distribution to a reflected Brownian motion:*

$$\sqrt{\frac{1-\rho}{\rho^2}}(Q(\rho^{-2}t) - \rho t) \Rightarrow B^+(t) \quad (1.8)$$

where $\Rightarrow$ indicates convergence in distribution, and $B^+(t)$ is the standard Brownian motion $B(t)$ reflected to remain non-negative.

Proof Idea: The result can be derived by considering the Lindley recursion for the waiting times and applying a functional central limit theorem after proper scaling and centering. The functional central limit theorem for the $M/M/1$ queue, leading to a reflected Brownian motion approximation in heavy traffic, can be found in:

Reference: In [18], Feller, W. (1971). *An introduction to probability theory and its applications (Vol. 2)*. John Wiley & Sons.

1.4.2 $M/G/1$ Queue

Theorem 1.4.2. *Consider an $M/G/1$ queueing system operating under a First-Come-First-Serve (FCFS) discipline with arrival rate λ . Let S represent the generic service time with $E[S] = 1/\mu$ and variance $\text{Var}(S) = \sigma^2$. Define the traffic intensity as $\rho = \lambda E[S]$. Let $W(t)$ denote the virtual waiting time at time t . If we assume that the second moment $E[S^2] < \infty$, then as $\rho \rightarrow 1$, the scaled and centered virtual waiting time process converges in distribution to a centered reflected Brownian motion:*

$$\sqrt{\frac{1-\rho}{\lambda\sigma^2}}(W(\rho^{-1}t) - (1-\rho)\rho^{-1}t) \Rightarrow W^*(t) \quad (1.9)$$

where $W^*(t)$ is a reflected Brownian motion with zero mean and variance σ^2 , reflected to stay non-negative.

Proof Idea: This result involves analyzing the $M/G/1$ waiting time recursion and then applying a functional central limit theorem, with appropriate scaling and centering, to the sequence of waiting times. The limiting process is the reflected Brownian motion due to the nature of the queue, ensuring that waiting times remain non-negative.

The functional central limit theorem and heavy traffic results for the $M/G/1$ queue can be found in:

Reference: In [30], Kingman, J. F. C. (1961). The single server queue in heavy traffic. *Proceedings of the Cambridge Philosophical Society*, 57(4), 902-904.

Additionally, Whitt's book provides a deeper dive into these approximations and more:

Reference: In [52], Whitt, W. (2002). *Stochastic-process limits*. Springer.

1.5 Reflected Brownian motion (RBM) as a main model of the limiting-process in the diffusion approximation

1.5.1 Functional Central Limit Theorems (FCLT) and the Rise of RBM in Queueing Systems

The Functional Central Limit Theorem, often referred to in the context of stochastic processes as the “invariance principle”, establishes the convergence of rescaled versions of partial sums of stochastic processes to a limiting continuous Gaussian process.

Theorem 1.5.1 (Donsker’s Invariance Principle). *Let $\{X_i\}$ be a sequence of independent and identically distributed random variables with $\mathbb{E}[X_i] = 0$ and $\text{Var}[X_i] = \sigma^2 < \infty$. Define the partial sum process $S_n(t) = \sum_{i=1}^{\lfloor nt \rfloor} X_i$. Then, as $n \rightarrow \infty$, the rescaled partial sum process*

$$W_n(t) = \frac{1}{\sqrt{n}} S_n(t)$$

converges in distribution in the Skorohod space $D[0, 1]$ (of right-continuous functions with left limits on $[0, 1]$) to a standard Brownian motion $B(t)$.

Proof Idea:

1. *Finite Dimensional Distributions:* Begin by proving that for any finite collection of times $0 \leq t_1 < t_2 < \dots < t_k \leq 1$, the joint distribution of $(W_n(t_1), W_n(t_2), \dots, W_n(t_k))$ converges to the joint distribution of $(B(t_1), B(t_2), \dots, B(t_k))$. This uses the classical Central Limit Theorem.
2. *Tightness:* To show convergence in the Skorohod space, it’s necessary to prove that the family of stochastic processes $\{W_n(t)\}$ is tight. This means that for any $\epsilon > 0$, there exists a compact set $K \subseteq D[0, 1]$ such that $\mathbb{P}(W_n \in K) > 1 - \epsilon$ for all n .

3. *Kolmogorov's Continuity Criterion*: Use Kolmogorov's criterion to establish the existence of a version of the Brownian motion that's continuous almost surely, which ensures convergence in $D[0, 1]$.
4. *Merging the Results*: Combining the results from the above steps gives the desired convergence in distribution in the Skorohod space $D[0, 1]$.

Reference:

Donsker, M. D. (1952). "Justification and Extension of Kolmogorov's Limit Theorems." *Proceedings of the National Academy of Sciences of the United States of America*, 38(3), 263-267.

Within the realm of queueing systems, this becomes particularly pertinent as we investigate the limiting behavior of queueing processes under suitable normalization, especially as the system is driven towards its capacity, commonly referred to as the "heavy traffic" regime.

To begin with, consider a queueing process $\{Q(t), t \geq 0\}$, where $Q(t)$ represents the queue length at time t . Under the usual conditions of stationarity and ergodicity, one can study the normalized centered and scaled process, $Q_n(t)$, defined as:

$$Q_n(t) = \frac{Q(nt) - n\mu t}{\sqrt{n}} \quad (1.10)$$

Here, μ represents the mean rate of arrivals (or service, depending on the context). The FCLT posits that, for a broad class of queueing models, $Q_n(t)$ converges weakly to a Gaussian process as $n \rightarrow \infty$.

However, the direct application of the FCLT might yield a Brownian motion which can take negative values, contradicting the physical reality of queues. This leads us to the realm of RBM. The Reflected Brownian Motion (RBM), defined as a Brownian motion reflected to remain non-negative, serves as the limiting process in these cases. Mathematically, if $B(t)$ is a standard Brownian motion, the RBM, $R(t)$, can be described as:

$$R(t) = B(t) + |\inf_{s \leq t} B(s)| \quad (1.11)$$

This reflection mechanism at the origin ensures that the process remains non-negative, thus preserving the intrinsic nature of queueing systems.

Furthermore, in systems where overloads are frequent, leading to long queues, the behavior of the queue length process, when suitably normalized, mimics the RBM. This is especially pronounced in systems operating under heavy traffic conditions, where the operational capacity closely matches the demand. The reflection at zero for the RBM effectively captures the system's inherent resilience in preventing queue lengths from becoming negative.

In conclusion, while the FCLT provides the foundational pathway to understand the limiting behaviors of queueing systems, it's the nuanced characteristics of RBM that align perfectly with the physical realities of queues, making it the primary candidate as the limiting process in diffusion approximations.

1.5.2 Reflected Brownian Motion: An Intrinsic Model for Queueing Systems

Within the domain of queueing theory, the Reflected Brownian Motion (RBM) and its variants have emerged as a frequently used model for the nuanced and multifaceted behavior of queueing systems, especially when venturing into the realms of their limiting behavior. This section meticulously examines the salient features of RBM that account for its primacy and resonance with real-world queueing dynamics.

1.5.2.1 Non-Negativity and Physical Constraints

The very nature of a queue precludes the possibility of negative lengths. Any model aspiring to be a representative of queueing dynamics needs to fundamentally respect this axiom. It's not merely about mathematical consistency, but more pertinently about physical

realism. Queues, in their essence, can either be empty or possess a certain positive length. The concept of a queue having a negative length is an anathema to both theoretical constructs and real-world observations.

While the Brownian motion does capture the inherent stochasticity, its trajectory isn't constrained to remain non-negative, making it an inappropriate representation for queue lengths. Here, RBM gains precedence. By introducing a reflection mechanism at the boundary (typically at zero), RBM ensures adherence to the non-negativity principle.

Formally, let $B(t)$ denote a standard Brownian motion. The Reflected Brownian Motion, $R(t)$, can be defined as:

$$R(t) = B(t) + \sup_{0 \leq s \leq t} \left(- \min_{0 \leq u \leq s} B(u) \right) \quad (1.12)$$

This mathematical formulation guarantees that the value of the process $R(t)$ remains non-negative at all times.

1.5.2.2 Stochastic Variability: Capturing Randomness in Arrivals and Services

Another quintessential aspect of queueing systems is the inherent randomness in the arrivals and service processes. While determinism might provide some average insights, it fails to capture the intricacies and volatilities of practical scenarios. Hence, there's an intrinsic need for a stochastic model.

Regular Brownian motion, with its inherent randomness, certainly comes close but falls short due to its aforementioned unboundedness. RBM, inheriting the stochastic properties of Brownian motion while being disciplined by its reflection mechanism, emerges as an optimal fit. The stochastic properties of RBM effectively encapsulate the randomness in arrivals and services, providing a holistic representation of the system.

1.5.2.3 Reflection Mechanism: A Mirror to Real-world Dynamics

Beyond mere mathematical convenience, the reflection property of RBM carries profound real-world implications. Consider the scenario of a server system nearing its capacity. The

buildup of the queue might lead to adaptive measures: increased service rates, activation of auxiliary servers, or even client redirection. Such mechanisms, designed to mitigate or prevent system overload, parallel the reflection mechanism intrinsic to RBM.

In a mathematical sense, when the process approaches a boundary (often zero in the context of queue lengths), the reflection ensures it remains within the permissible domain. In real-world analogies, this reflection can be likened to systemic responses invoked to manage and maintain service quality and efficiency.

In essence, the RBM's alignment with the core principles of queueing systems isn't coincidental. It emerges as an outcome of an intricate interplay of mathematical formulations and their resonances with real-world dynamics, making RBM an unparalleled model in the domain of queueing theory.

1.5.3 Insights and Implications from RBM-based Models

The salient attributes of Reflected Brownian Motion (RBM) in elucidating the nuanced intricacies of queueing systems cannot be overstated. While its descriptive power serves as a foundational scaffold, its analytical prowess opens the avenue for myriad insights and actionable stratagems. This section delves deep into these implications, intertwining them with mathematical formalisms.

1.5.3.1 System Dynamics and Behavioral Analysis

The RBM model's fidelity to realistic queueing dynamics offers an unparalleled lens into the intricate mechanics of such systems, especially in the vicinity of their operational thresholds.

1. Queue Variability and Stochastic Analysis: The RBM, represented as

$$R(t) = B(t) + \sup_{0 \leq s \leq t} \left(- \min_{0 \leq u \leq s} B(u) \right),$$

captures the stochastic variability of the queue lengths. The stochastic properties such as variance and autocorrelation can provide insights into traffic predictability. Peaks in variance, for instance, can signal potential system bottlenecks. See [57] for reference.

2. **Threshold Behavior and System Overflows:** From the reflection property, one can derive thresholds, say T , beyond which the system experiences overflows. Mathematically, the reflection point offers a pivotal insight:

$$P(R(t) \geq T),$$

where P denotes the probability. A higher value could be an indicator of frequent overflows, necessitating infrastructural enhancements or adaptive protocols. For more details, see [29].

1.5.3.2 Resource Management and Optimization

With the granular insights furnished by RBM, system architects and administrators are equipped with a potent toolkit for resource orchestration.

1. **Dynamic Allocation and Stochastic Control:** Based on the trajectory of the RBM, one can devise a stochastic control strategy $u(t)$ that modulates resource allocation in real-time. The objective could be to minimize the expected queue length, formulated as:

$$\mathbb{E}\left[\int_0^T R(t)u(t)dt\right],$$

where $\mathbb{E}$ denotes expectation and T is a suitable time horizon. For more details, see [24].

2. **Optimal Utilization and Performance Analysis:** Through the rigorous analysis of the RBM, optimal strategies $u^*(t)$ can be discerned, ensuring resources are optimally utilized. This minimizes operational costs while maximizing throughput, a balance

given by:

$$\min_{u(t)} \mathbb{E} \left[\int_0^T c(u(t)) dt + h(R(T)) \right],$$

where c represents cost and h encapsulates a suitable performance metric. Further details can be found in [5]

3. **Metrics Derivation:** Metrics such as mean waiting time W , throughput λ , and service rate μ can be derived from the RBM model. By analyzing the steady-state properties, one can express:

$$W = \frac{L}{\lambda},$$

where L represents the mean number of customers in the system, yielding insights into system efficiency and areas of enhancement. See [43] for further details.

To encapsulate, the probing into Reflected Brownian Motion as a modeling paradigm for queueing systems isn't only a pure academic exercise. It has been spread over academic research (both theoretical and applied), engineering, industrial and a lot more, each mathematical twist and turn shedding light on profound and actionable insights with a vast spectrum of real-world ramifications.

Chapter 2

Reflected Brownian motion and its formulations

2.1 1-dim RBM on the positive half of real line

Reflected Brownian Motion in one dimension is a cornerstone in stochastic process analysis, given its wide applicability and profound theoretical importance.

2.1.1 An explicit pathwise formula of 1-Dimensional RBM

A particularly good character of 1-dim RBM is that it possesses an explicit path wise formula due to the simplicity of the structure of its state-space.

RBM in one dimension is denoted by $Z(t)$ and is defined as:

$$Z(t) = B(t) + \sup_{0 \leq s \leq t} \left(- \min_{0 \leq u \leq s} B(u) \right), \quad (2.1)$$

where $B(t)$ represents a standard Brownian motion.

2.1.2 1-dim Skorokhod Problem

The Skorokhod problem provides a formulation for the reflection mechanism in RBM. Given a continuous function $x(t)$, the problem seeks a pair of functions $(z(t), l(t))$ such that:

$$z(t) = x(t) + l(t) \geq 0, \quad (2.2)$$

$$l(t) \text{ is non-decreasing,} \quad (2.3)$$

$$l'(t) = 0 \text{ wherever } y(t) > 0. \quad (2.4)$$

2.1.3 Stationary Distribution

For a 1-dimensional RBM Z_t , under conditions when the driving Brownian motion has a drift μ toward the origin and variance σ^2 , (i.e. negative drift for RBM on the positive half real line) the stationary distribution is:

$$\mathbb{P}(Z_t \leq x) = 1 - \exp \left\{ \frac{2\mu x}{\sigma^2} \right\}, \text{ for } \mu < 0. \quad (2.5)$$

See Harrison [21] page 15 for details. It should be pointed out that in Harrison [21], the 1-dim has been investigated very extensively, but in another name called “regulated Brownian motion”.

2.1.4 Tanaka’s Formula

2.1.4.1 Local Time

For a one-dimensional Brownian motion B_t , the local time L_t^a at level a up to time t is defined as follows:

$$L_t^a = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{2\varepsilon} \int_0^t \mathbf{1}_{\{|B_s - a| < \varepsilon\}} ds.$$

Intuitively, this measures the “amount of time” the Brownian motion has spent around the level a up to time t , accumulating whenever the process is near a .

2.1.4.2 The theorem of Tanaka's Formula

Let B_t be a one-dimensional standard Brownian motion defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then, for all $t \geq 0$:

$$|B_t| = B_0 + \int_0^t \text{sgn}(B_s) dB_s + \frac{1}{2}L_t^0.$$

Here, sgn denotes the sign function, defined by:

$$\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{if } x = 0, \\ -1 & \text{if } x < 0. \end{cases}$$

Proof

The proof utilizes Itô's formula for semimartingales. We aim to apply Itô's formula to the function $f(x) = |x|$. The challenge is that f is not differentiable at 0, but a careful treatment of the terms yields the result.

1. For $x \neq 0$, $f'(x) = \text{sgn}(x)$. For $x = 0$, we'll define $f'(0) = 0$, keeping in mind the derivative does not actually exist at this point.
2. Applying Itô's formula to f and considering the properties of the quadratic variation of the Brownian motion:

$$df(B_t) = f'(B_t) dB_t + \frac{1}{2}f''(B_t) d[B]_t.$$

Given that $f''(B_t)$ does not exist, the second term poses an issue. However, due to the behavior of the Brownian motion, this term is only problematic where $B_t = 0$.

3. Observing the behavior of B_t around 0, the local time L_t^0 accounts for the “missing” term in the Itô's formula application, giving the Tanaka's formula in its entirety.

2.2 RBM in an Orthant

RBM in an orthant is the roughly speaking the second complicated case than RBM on the positive half of real line. However, being in an orthant ($\dim \geq 2$) is still essentially make the problem go to another essentially more complicated level. We summarize the seminal results among many exciting ones here, to show the monuments of the reserach on this topic, as well as some possible futuristic research problems and concerns

2.2.1 RBM in an Orthant: Fundamental Results

1. **Existence and Uniqueness:** There exists a unique solution to the Skorokhod problem in the orthant, corresponding to RBM.

Reference: Harrison, Reiman (1981) [22].

2. **Stationary Distributions:** Under specific conditions, RBM in an orthant admits a stationary distribution. The existence of such a distribution provides a foundation for many applications and further theoretical advancements.

Reference: Williams (1987) [55].

3. **Decomposition Principle:** Certain network structures allow RBM in an orthant to be decomposed into simpler, lower-dimensional RBMs.

Reference: Dai (1995) [10].

2.2.2 Applications and Extensions

1. **Queueing Networks:** RBM in orthants can be utilized to model and analyze queueing networks, especially in heavy traffic scenarios.

Reference: Harrison (1987) [24].

2. **Stochastic Control & Optimization:** RBMs can be controlled to optimize performance measures. This area merges control theory, optimization, and stochastic

processes.

Reference: Sznitman (1991) [49].

3. **Higher-Dimensional Analysis:** As systems become more intricate, there's been a surge in interest in analyzing RBM behavior in higher-dimensional orthants, especially concerning corner reflections.

Reference: Dupuis and Williams (1994) [16].

2.2.3 Trends on the reserach and Challenges (as of 2023)

1. **Scaling and Computational Aspects:** With the rise of large-scale systems and networks, there's a growing emphasis on the computational aspects of RBMs. Numerical methods, simulations, and approximations are areas of active research.
2. **Interplay with Modern Domains:** There's a push towards integrating the insights from RBM research with modern domains like machine learning, data centers, and cloud computing.
3. **Multi-class Networks:** Systems with multiple classes of jobs/customers/agents present a richer structure and complexity. This has become a focal point of recent investigations.

2.2.4 Open Problems and Challenges (as of 2023)

1. **Generalized Reflections:** While the behavior of RBM at boundaries is well-understood, how RBM behaves at the corners of higher-dimensional orthants, especially with non-standard reflection laws, remains an area to delve into.
2. **Transient Behavior:** Much of the existing work has focused on stationary behavior. Understanding the transient behavior of RBMs, especially in complex networks, remains challenging.

3. **Integration with Modern Statistical Methods:** With the avalanche of data in many applications, integrating RBM models with statistical methods to estimate model parameters and validate theoretical results is a growing need.
4. **Non-Markovian Models:** Extending the analysis of RBM to scenarios where the underlying driving processes are not Markovian is an ambitious and challenging frontier.

2.3 Skorokhod problem

The Skorokhod problem, which seeks regulated functions constrained by given barriers, has found significant applicability in studying reflected Brownian motion (RBM) [44]. By providing a robust framework for understanding reflection mechanisms, it allows for a deeper exploration of RBM's behavior at boundaries [21]. The multidimensional extensions further elucidate the complexities of RBM in higher-dimensional spaces [16]. This relationship between the Skorokhod problem and RBM serves as a bedrock in applications from queueing theory to advanced computational systems [55].

2.3.1 Skorokhod's Original Problem and Solution

In the **Paper** [45], by Skorokhod, A. V. (1956), "Limit theorems for stochastic processes". He first introduced a space of regulated functions, and posed the problem of finding a regulated version of a function that remains greater than a fixed barrier. And then he Provided conditions under which solutions exist. This is the first time the Skorokhod problem showed up.

2.3.2 Reflected Brownian Motion in an Orthant

In the **Paper** [23], by Harrison, J. M., and Reiman, M. I. (1981), "Reflected Brownian motion on an orthant", they introduced and rigorously formalized RBM in multidimensional

spaces, and demonstrated existence and uniqueness of RBM in a non-negative orthant, and then showed that this RBM is a strong Markov process.

2.3.3 Multidimensional RBM and Stationary Distributions

In the **Paper** [25], by Harrison, J. M., and Williams, R. J. (1987). “Multidimensional reflected Brownian motions having exponential stationary distributions.” They studied RBM in multiple dimensions, derived conditions for RBM in an orthant to have exponential stationary distributions, and examined the construction and properties of these distributions.

2.3.4 The Skorokhod Problem in the Quadrant

In the **Paper** [15], by Dupuis, P., and Ishii, H. (1991), “On Lipschitz continuity of the solution mapping to the Skorokhod problem, with applications.” They investigated the Skorokhod problem in the two-dimensional quadrant, and explored Lipschitz continuity of the solution to the Skorokhod problem and conditions for its existence, and showed some applications of these results to stochastic differential equations with RBM.

2.3.5 Skorokhod Problem in smooth and non-smooth bounded Domains

In the **Paper** [33] by Paper: Lions, P.-L., and Sznitman, A.-S. (1984). “Stochastic Differential Equations with Reflecting Boundary Conditions.” They developed a systematic framework for analyzing stochastic differential equations with reflecting boundary conditions, particularly in bounded Lipschitz domains, presented a new approach to solving the Skorokhod problem, which provided clearer insights into the structure and properties of solutions, and showed the strong connections between their framework and the behavior of RBM, establishing the foundation for further studies on reflected diffusions in various geometric domains, by using penalization methods.

2.3.6 Skorokhod Problem in General Domains

In the **Paper** [42] by Saisho, Y. (1987), “Stochastic differential equations for multi-dimensional domain with reflecting boundary.” They extended Skorokhod problem to more general domains, and investigated the existence, uniqueness, and properties of solutions in these contexts, and then Demonstrated connections between RBM and the Skorokhod problem in these broader settings.

2.4 Submartingale problem

The first place where the submartingale problem shows up is the paper by Stroock and Varadhan [46]. In that paper, they formulated the submartingale problem for “diffusion processes with boundary conditions in a smooth domain” as an analogue to the martingale problem for “diffusion processes” [48] and [47].

The idea behind the submartingale problem is to characterize a diffusion associated to a second-order elliptic operator in some domain with boundary conditions, by a unique measure on the space of continuous functions (i.e. the space of sample paths of diffusion processes), which satisfies an initial condition and makes an integral functional of the underlying diffusion process a submartingale.

To this end, a diffusion process getting reflected in a constrained domain is characterized in an intrinsic way. By intrinsic, we mean that we don’t need a boundary process which “pushes” the diffusion process back into the domain when it gets to the boundary, noticing that the boundary process has a different state space other than the state space of the diffusion process its own. Although, in most of the cases, such a boundary process can be proved to exist in a hindsight way.

However, the form of submartingale problem in our concern is from the paper by Varadhan and Williams [51], a submartingale problem in a more singular domain, the wedge-shaped area.

Definition 2 (Submartingale Problem). *A family of probability measures $\{\mathbb{P}^z, z \in S\}$ on $(C_S, \mathcal{M})$ is said to solve the submartingale problem if for each $z \in S$, the following three conditions hold,*

1. $\mathbb{P}^z(Z(0) = z) = 1;$

2. *For each $f \in C_b^2(S)$, the process*

$$\left\{ f(Z(t)) - \frac{1}{2} \int_0^t \Delta f(Z(s)) ds, t \geq 0 \right\}$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}^z)$ whenever f is constant in a neighbourhood of the origin and satisfies $D_i f \geq 0$ on ∂S_i for $i = 1, 2;$

3. $\mathbb{E}_\mu^z \left[\int_0^\infty \mathbf{1}_{\{Z(t)=0\}} dt \right] = 0.$

Chapter 3

Introducing the underlying research problem

To set up the research problem, we study 2-dimensional Brownian motion with constant drift $\mu \in \mathbb{R}^2$ constrained to a wedge S in $\mathbb{R}^2$. This process may also be referred to as reflected Brownian motion (RBM) with drift in a wedge, and we denote the process itself by Z . For concreteness, we define the wedge in polar coordinates by $\{r \geq 0, 0 \leq \theta \leq \xi\}$ for some $0 < \xi < 2\pi$. Loosely speaking, the behavior of Z may be characterized as follows. In the interior of S , Z behaves as a 2-dimensional Brownian motion. On the other hand, the behavior of Z on the boundary of S is characterized by two reflection angles θ_1 and θ_2 , depending upon whether the lower boundary ∂S_1 or upper boundary ∂S_2 has been reached. Both $-\pi/2 < \theta_1, \theta_2 < \pi/2$ and the angles are measured from their inward-facing normals, with positive angles corresponding to reflection toward the vertex of the wedge and negative angles away. See Figure below for an illustration.

One way to define RBM in a wedge is using a sample-path approach [9, 11, 23, 50] where Z is explicitly characterized as the sum of a 2-dimensional Brownian motion on an arbitrary probability space [28, 32, 40] and a constraining or pushing process which satisfies the specifications related to the directions of reflection given above. This sample-path approach

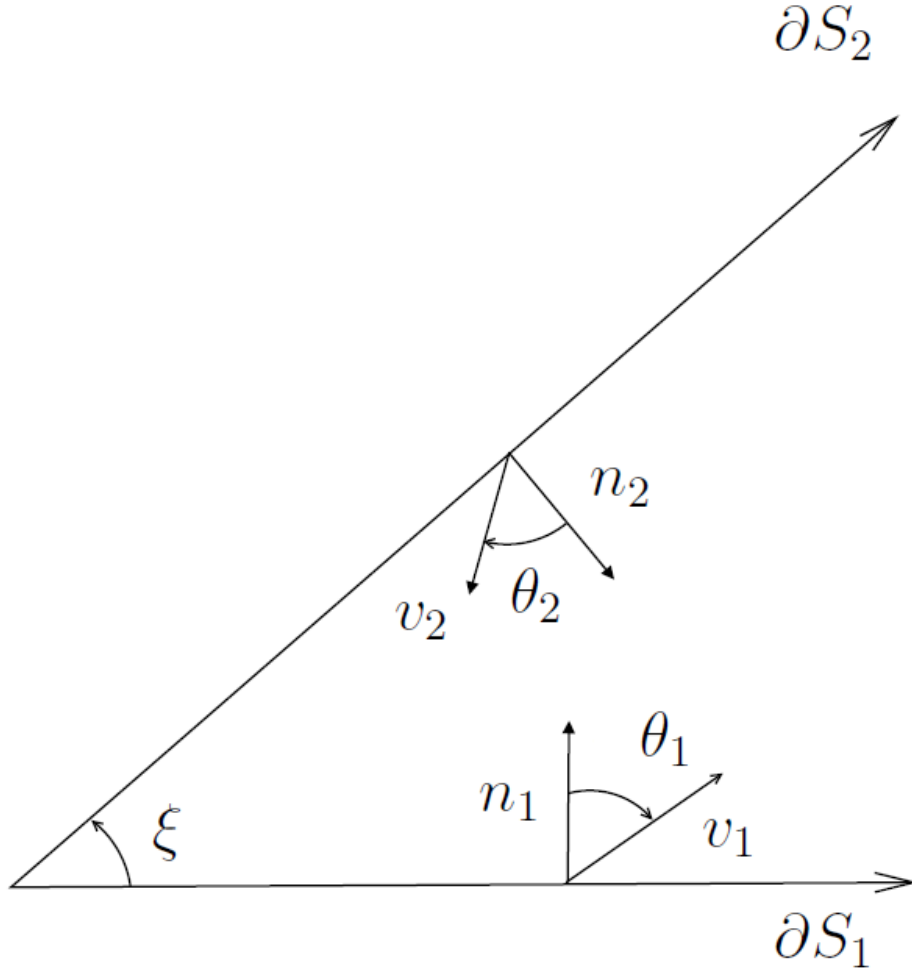


Figure 3.1: RBM in a Wedge

works with or without drift for some but not all parameter regimes of $(\xi, \theta_1, \theta_2)$. It tends not to work in regimes where Z is known not to be a semi-martingale [54] and the pushing process has infinite variation. Recent progress in this direction has however been made [26, 37].

A more probabilistic approach to defining RBM in a wedge was given by Varadhan and Williams [51]. In this case, Z is defined as the solution to a submartingale problem. This approach yields existence and uniqueness results for all parameter regimes but at several points the proofs of [51] rely heavily on the assumption that Z behaves as a standard Brownian motion inside of S . This is not an issue for parameter regimes where the sample-path approach described above may be applied because it is amenable to Brownian motions with

drift, and the recent paper [27] demonstrates equivalence between the sample-path and the submartingale approach in such settings. On the other hand, in parameter regimes where the sample-path approach cannot be applied, extending the results of [51] in the direction of allowing Z to behave as a Brownian motion with drift in the interior of S remains an open problem. In this paper, we resolve this open problem for the case of a constant drift.

Remark 3.0.1. We conjecture that our results could be further generalized for the case when the drift is a bounded function of the current state, but this generalization is beyond the scope of the present paper.

As introduced in Chapter 1, the primary motivation comes from queueing theory where semi-martingale RBM with drift has long been known to serve as the weak limit of both the properly scaled queue length [4, 20, 22, 39] and workload [3, 6, 35, 56] processes of different queueing systems in heavy-traffic. In such queueing settings, the drift term arises as the result of an imbalance between the input and output processes to the system. The limiting RBM in these cases is often defined using the sample-path approach via the conventional Skorokhod map [23, 44, 52]. More recently, using the extended Skorokhod map [37], RBM with drift which is not a semi-martingale has been proven [38] to be the weak limit of the properly scaled unfinished work process of the generalized processor sharing model in heavy traffic. In this example, the sample-path approach is still employed to define the limiting process with the help of the extended Skorokhod map [37]. We conjecture however that there exist other applied queueing settings where the limiting heavy-traffic process is an RBM with drift which is not a semi-martingale and cannot be rigorously defined via the sample-path approach. One of these settings is the coupled processor model [7, 17]. In such situations, before proving any limit theorems, it is necessary to first establish the existence of RBM with drift through other means such as the submartingale problem.

Remark 3.0.2. We also mention that there exists a related stream of literature studying the behavior of reflected Brownian in smooth domains with cusps. The paper [13] appears to

be the first to study reflected Brownian motion confined to a cusp in the plane. In this case, RBM is defined as the solution to a corresponding submartingale problem. Existence and uniqueness results are then proven by conformally mapping RBM in the upper half plane to the cusp and applying a time change. In the follow-up paper [12], it is shown that depending on the geometry of the problem in [13], RBM in a cusp in the plane may or may not turn out to be a semi-martingale. The results in [12] are similar to those in [54] where a wedge instead of a cusp is considered. The authors in [19] use a Dirichlet form approach to construct a diffusion process contained in a d -dimensional Lipschitz domain with cusps. Moreover, conditions are provided under which the constraining process is of bounded variation. The paper [8] considers RBM in a cusp in the plane as the solution to a stochastic differential equation with reflection (SDER). It is proven that in this case there exists a unique weak solution to the corresponding SDER.

To study RBM, this thesis focuses on the submartingale problem with drift in a wedge, and its associated absorption problem and hitting probabilities. We establish the existence and uniqueness of the submartingale problem with drift, and Markov and Feller properties. The formulation of the research problem is stated as follows.

We first set up some notation to make our further statement clear.

Let $C_S = C(\mathbb{R}_+, S)$ and, for each $t \geq 0$, let $Z(t) : C_S \rightarrow S$ denote the coordinate map $Z(t)(\omega) = \omega(t)$ for $\omega \in C_S$. Also, let $Z = \{Z(t), t \geq 0\}$ denote the coordinate mapping process on C_S . Let $\mathcal{M}_t = \sigma(Z(s), 0 \leq s \leq t)$ be the underlying natural filtration with terminal σ -algebra $\mathcal{M} = \sigma(Z(s), s \geq 0)$. For each $n \geq 1$ and domain $\Omega \subseteq \mathbb{R}^2$, we denote by $C_b^n(\Omega)$ the set of n times bounded continuously differentiable functions on Ω . We assume that the wedge S is positioned so that one side of it is the positive horizontal half line, and the angle of the wedge is ξ . We define ∂S_1 and ∂S_2 as the two sides of the wedge so that neither includes the vertex, i.e., $\partial S_1 = \{(x, 0) : x > 0\}$ and $\partial S_2 = \{r(\cos \xi, \sin \xi) : r > 0\}$. Next (see Figure 3.1), we denote by v_1 and v_2 the reflection directions on the boundaries ∂S_1 and ∂S_2 , respectively. For convenience, we assume that each v_i is normalized such that $v_i \cdot n_i = 1$,

where n_i is the inward facing normal vector on ∂S_i for $i = 1, 2$. Finally, for $i = 1, 2$, we set the directional derivative operator $D_i = v_i \cdot \nabla$, with ∇ being the gradient operator, the dot is the inner product, and denote by Δ the Laplacian operator.

3.1 The Absorbed Process Problem with Drift

Let

$$\tau_0 = \inf\{t \geq 0 : Z(t) = 0\}$$

be the stopping time with respect to $\{\mathcal{M}_t, t \geq 0\}$ representing the first time that Z reaches the vertex of the wedge. Results in this subsection concern the RBM in a wedge up until τ_0 . Results of this type were provided in [51] for the driftless case and we extend many of them to the case of a constant drift μ . We begin with the following definition.

Definition 3 (The Absorbed Process Problem). *A family of probability measures $\{\mathbb{P}_\mu^{z,0}, z \in S\}$ on $(C_S, \mathcal{M})$ is said to solve the absorbed process problem with drift $\mu \in \mathbb{R}^2$ if for each $z \in S$, the following three conditions hold,*

1. $\mathbb{P}_\mu^{z,0}(Z(0) = z) = 1;$

2. *The process*

$$\left\{ f(Z(t \wedge \tau_0)) - \int_0^{t \wedge \tau_0} \mu \cdot \nabla f(Z(s)) ds - \frac{1}{2} \int_0^{t \wedge \tau_0} \Delta f(Z(s)) ds, t \geq 0 \right\}$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^{z,0})$, for each $f \in C_b^2(S)$ such that $D_i f \geq 0$ on ∂S_i for $i = 1, 2$;

3. $\mathbb{P}_\mu^{z,0}(Z(t) = 0, \forall t \geq \tau_0) = 1.$

Problem 3.1.1. For the above absorbed process problem with drift,

1. if there is a solution to this problem, is it unique?

2. once such a solution to this problem exists, what is the hitting probability to the origin (the vertex of the wedge)?

3.2 The Submartingale Problem with Drift

Definition 4 (Submartingale Problem with Drift). *A family of probability measures $\{\mathbb{P}_\mu^z, z \in S\}$ on $(C_S, \mathcal{M})$ is said to solve the submartingale problem with drift $\mu \in \mathbb{R}^2$ if for each $z \in S$, the following three conditions hold,*

1. $\mathbb{P}_\mu^z(Z(0) = z) = 1;$
2. For each $f \in C_b^2(S)$, the process

$$\left\{ f(Z(t)) - \int_0^t \mu \cdot \nabla f(Z(s)) ds - \frac{1}{2} \int_0^t \Delta f(Z(s)) ds, t \geq 0 \right\}$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ whenever f is constant in a neighbourhood of the origin and satisfies $D_i f \geq 0$ on ∂S_i for $i = 1, 2$;

3. $\mathbb{E}_\mu^z \left[\int_0^\infty \mathbf{1}_{\{Z(t)=0\}} dt \right] = 0.$

Problem 3.2.1. For the above submartingale problem with drift,

1. if there is a solution to this problem, is it unique?
2. once such a solution to this problem exists, does it have Markov property and Feller property and other probabilistic properties?

Chapter 4

Absorption problem and hitting probability

4.1 The problem and the main results

Recall from Chapter 3 that

$$\tau_0 = \inf\{t \geq 0 : Z(t) = 0\}$$

is the stopping time with respect to $\{\mathcal{M}_t, t \geq 0\}$ representing the first time that Z reaches the vertex of the wedge.

Results in this section concern the RBM in a wedge up until τ_0 . Results of this type were provided in [51] for the driftless case, and we extend many of them to the case of a constant drift μ . We begin with the following definition. For convenience, we display the figure of the state space here again.

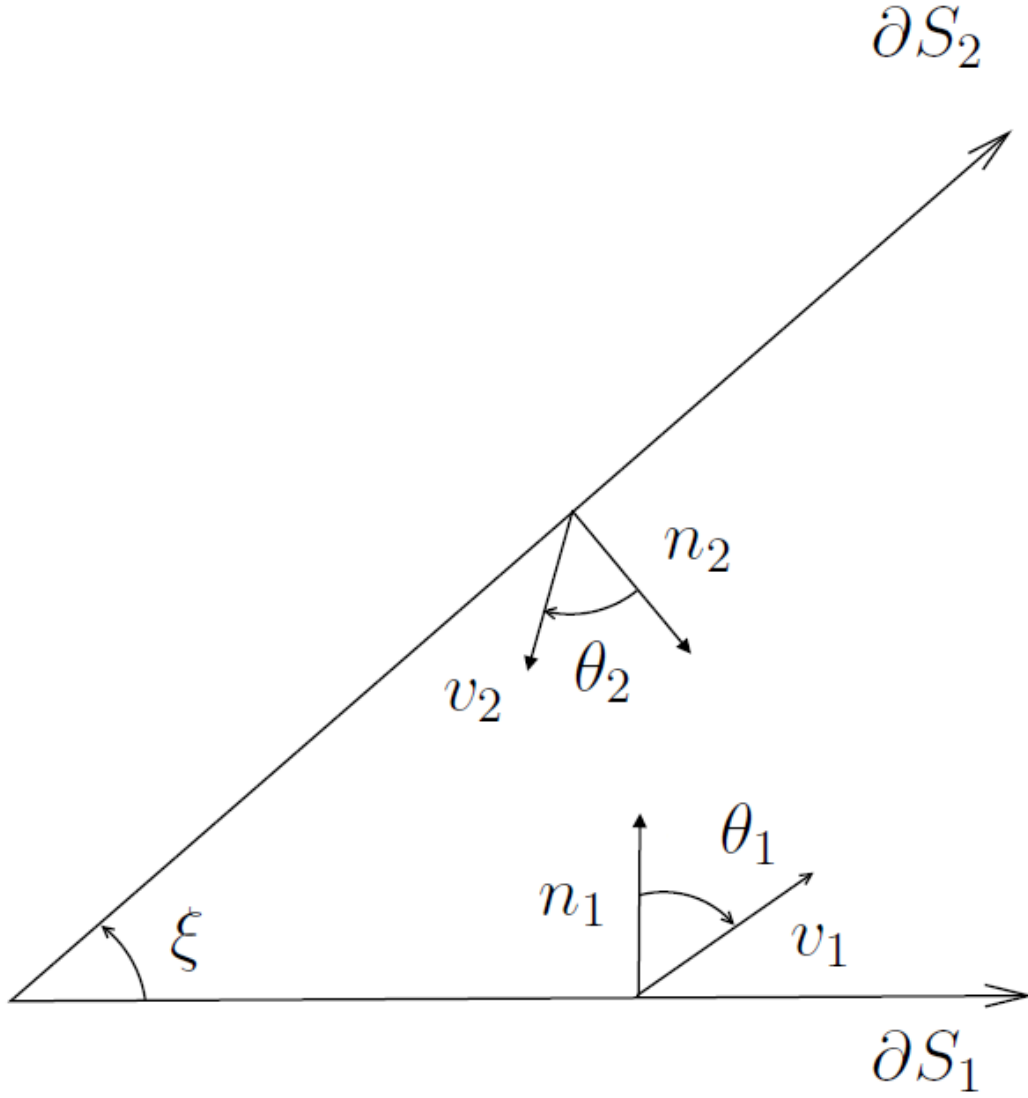


Figure 4.1: RBM in a wedge

Definition 5 (The Absorbed Process Problem). *A family of probability measures $\{\mathbb{P}_\mu^{z,0}, z \in S\}$ on $(C_S, \mathcal{M})$ is said to solve the absorbed process problem with drift $\mu \in \mathbb{R}^2$ if for each $z \in S$, the following three conditions hold:*

1. $\mathbb{P}_\mu^{z,0}(Z(0) = z) = 1$;
2. The process

$$\left\{ f(Z(t \wedge \tau_0)) - \int_0^{t \wedge \tau_0} \mu \cdot \nabla f(Z(s)) ds - \frac{1}{2} \int_0^{t \wedge \tau_0} \Delta f(Z(s)) ds, t \geq 0 \right\}$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^{z,0})$, for each $f \in C_b^2(S)$ such that $D_i f \geq 0$ on ∂S_i for $i = 1, 2$;

3. $\mathbb{P}_\mu^{z,0}(Z(t) = 0, \forall t \geq \tau_0) = 1$.

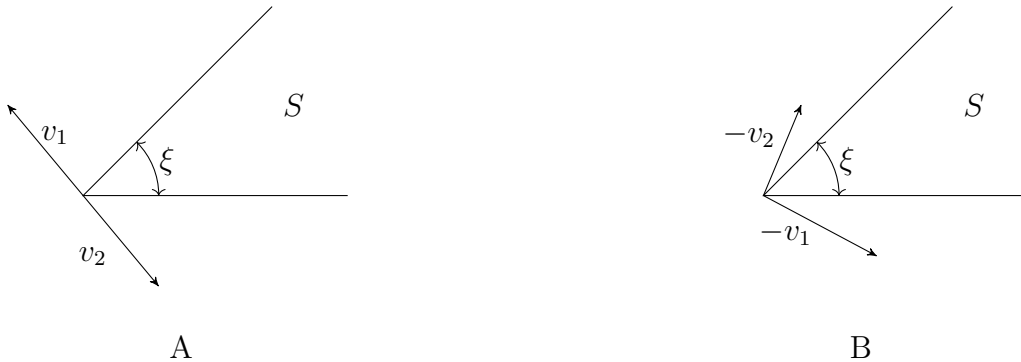
Theorem 4.1.1. *For each $\mu \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$, there exists a unique solution to the absorbed process problem.*

The above theorem is particularly interesting if $\alpha \geq 2$, since Theorem 5.2.1 does not cover that case. The existence of a solution to the absorbed process problem easily follows from the existence of a solution to the submartingale problem whenever $\alpha < 2$. However, the uniqueness part of Theorem 4.1.1 does not follow in an obvious way from Theorem 5.2.1, even in the $\alpha < 2$ case. Our proof of Theorem 4.1.1 applies to all $\alpha \in \mathbb{R}$.

Next, we state a series of results on the hitting probability of the vertex for the absorbed process in the case of a constant drift.

Theorem 4.1.2. *If $\alpha \leq 0$, then for each $\mu \in \mathbb{R}^2$ and $z \in S$, $\mathbb{P}_\mu^{z,0}(\tau_0 = \infty) = 1$.*

The hitting probability of the vertex is more varied in the case of $\alpha \geq 1$, and before proceeding we must make some observations on the geometry of the wedge. For $n \geq 1$ and a set of vectors $\{a_1, \dots, a_n\} \subset \mathbb{R}^2$, let $\text{co}(a_1, \dots, a_n)$ denote the closed convex cone generated by $\{a_1, \dots, a_n\}$. We illustrate two relevant cases for α below.



In the above diagrams, case A corresponds to $\alpha = 1$, which occurs if and only if $\text{co}(v_1, v_2)$ is a line. Case B corresponds to $\alpha > 1$, which occurs if and only if $\text{co}(-v_1, -v_2)$ contains S .

In both cases, that is, whenever $\alpha \geq 1$, we have $\text{co}(v_1, v_2) \cap S = \{0\}$. Note also that $\alpha \geq 1$ implies $\xi < \pi$.

Theorem 4.1.3. *If $\alpha \geq 1$, then*

$$\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) > 0 \text{ for each } \mu \in \mathbb{R}^2 \text{ and } z \in S. \quad (4.1)$$

Moreover, if in addition to the $\alpha \geq 1$ condition we also have that

$$\text{co}(v_1, v_2, \mu) \cap S = \{0\}, \quad (4.2)$$

then for each $z \in S$,

$$\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) = 1. \quad (4.3)$$

We note that in the case $\alpha > 1$, condition (4.2) can be cast in an algebraic form. Let R be the 2×2 matrix such that its i -th column vector is v_i , for $i = 1, 2$. If $\alpha > 1$, then condition (4.2) is equivalent to the requirement that the vector $R^{-1}\mu$ has at least one non-negative component.

Remark 4.1.4. Theorem 4.1.3 leaves open the possibility that $\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) = 1$ whenever $\alpha \geq 1$. This, however, is not the case, as the following counterexample shows. Let $\alpha \in \mathbb{R}$ be arbitrary and let the drift $\mu \in \mathbb{R}^2$ be given by $\mu = \|\mu\|(\cos \eta, \sin \eta) \neq 0$, where $\eta \in (0, \xi)$. Then, it is not hard to show using the proposition below that $\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) < 1$ for each $z \in S \setminus \{0\}$.

Proposition 4.1.5. *Let S be the 2-d wedge defined above, let S^0 be the interior of S , let B be a 2-d standard Brownian motion with zero drift started at the origin under a probability measure P , and let $\mu \in \mathbb{R}^2$ be given by $\mu = \|\mu\|(\cos \eta, \sin \eta) \neq 0$, where $\eta \in (0, \xi)$. Then, if $0 < \xi < \pi$, for each $z \in S^0$,*

$$P(z + B_t + \mu t \in S^0, t \geq 0) > 0.$$

Using the proposition above and Theorem 4.1.3, we may now deduce that if $\alpha \geq 1$, $\eta \in (0, \xi)$, and $\mu = \|\mu\|(\cos \eta, \sin \eta) \neq 0$, then

$$\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) \in (0, 1) \quad \text{for each } z \in S \setminus \{0\}.$$

This implies that when $\alpha \geq 1$, hitting the vertex is no longer a 0-1 event for certain values of μ , which contrasts with the driftless result of [51].

4.2 Proof of Theorems 4.1.2, 4.1.3, and Proposition 4.1.5

We need to recall Proposition 5.6.1. According to this proposition, for every $w \in C(\mathbb{R}_+, \mathbb{R}^2)$ with $w(0) \in S$ there exists a triple (ϕ, η, T_0) such that items 1–4 and (i), (ii) hold. Since B is the coordinate mapping process on $C(\mathbb{R}_+, \mathbb{R}^2)$, we may replace w in item 1 with B , and write

$$\phi(t) = B(t) + R\eta(t), \quad t \in [0, T_0).$$

We also know that B is a two-dimensional Brownian motion with drift μ started at z under $\hat{\mathbb{P}}_\mu^z$ for every $z \in S$, hence we can write

$$\phi(t) = z + W(t) + \mu t + R\eta(t), \quad t \in [0, T_0), \tag{4.4}$$

where W is a standard two-dimensional Brownian motion started at zero under $\hat{\mathbb{P}}_\mu^z$.

The measure $\mathbb{P}_\mu^{z,0}$ was defined as the measure induced by Γ on $\mathcal{M}$ under $\hat{\mathbb{P}}_\mu^z$. From this and from $\tau_0 \circ \Gamma = T_0$ it follows that

$$\mathbb{P}_\mu^{z,0}(\tau_0 < \infty) = \hat{\mathbb{P}}_\mu^z(T_0 < \infty). \tag{4.5}$$

We shall use (4.5) repeatedly in the proofs below.

Proof of Theorem 4.1.2. By Theorem 2.2 in [51], we have $\mathbb{P}_0^{z,0}(\tau_0 = \infty) = 1$, thus by (4.5) also $\hat{\mathbb{P}}_0^z(T_0 = \infty) = 1$. For every $n \in \mathbb{N}_+$, the measures $\hat{\mathbb{P}}_0^z$ and $\hat{\mathbb{P}}_\mu^z$ are mutually absolutely continuous on $\mathcal{W}_n$, so $\hat{\mathbb{P}}_0^z(T_0 < n) = 0$ implies $\hat{\mathbb{P}}_\mu^z(T_0 < n) = 0$. Then $\hat{\mathbb{P}}_\mu^z(T_0 = \infty) = 1$ follows, and this together with (4.5) gives the required result. $\square$

Proof of Theorem 4.1.3. First, we are going to show (4.1). By the $\alpha \geq 1$ condition there exists a vector $b \in \mathbb{R}^2$ such that $b \cdot z < 0$ for all $z \in S$, $z \neq 0$, and $b \cdot v_i \geq 0$ for $i = 1, 2$. Indeed, if $\alpha \geq 1$, then $\text{co}(-v_1, -v_2)$ is either a line containing S within one side, or it is a wedge with an angle less than π containing S . In either case, the existence of such a vector follows. Then, by identity (4.4), for each $z \in S$,

$$0 \geq b \cdot \phi(t) = b \cdot z + b \cdot W_t + b \cdot v_1 \eta_1(t) + b \cdot v_2 \eta_2(t) + b \cdot \mu t \geq b \cdot z + b \cdot W_t + b \cdot \mu t,$$

for $t < T_0$, $\hat{\mathbb{P}}_\mu^z$ -a.s., and so

$$\hat{\mathbb{P}}_\mu^z(0 \geq b \cdot z + b \cdot W_t + b \cdot \mu t, t < T_0) = 1.$$

Therefore,

$$\begin{aligned} & \hat{\mathbb{P}}_\mu^z(0 \geq b \cdot z + b \cdot W_t + b \cdot \mu t, t < \infty) \\ & \geq \hat{\mathbb{P}}_\mu^z(\{0 \geq b \cdot z + b \cdot W_t + b \cdot \mu t, t < T_0\} \cap \{T_0 = \infty\}) \\ & = \hat{\mathbb{P}}_\mu^z(T_0 = \infty). \end{aligned}$$

This implies

$$\hat{\mathbb{P}}_\mu^z(-b \cdot z \geq b \cdot W_t + b \cdot \mu t, t < \infty) \geq \hat{\mathbb{P}}_\mu^z(T_0 = \infty). \quad (4.6)$$

However,

$$\hat{\mathbb{P}}_\mu^z(-b \cdot z \geq b \cdot W_t + b \cdot \mu t, t < \infty) < 1,$$

and together with (4.5) this proves the result.

Suppose now that (4.2) also holds, in addition to $\alpha \geq 1$. Then $\text{co}(v_1, v_2, \mu)$ is either a wedge with an angle less than π , or a half-space, or a line. Then the same is true for $\text{co}(-v_1, -v_2, -\mu)$, and if it is a wedge or a half-space then it contains S , and if it is a line then it contains S in one side. In all cases, we can select b so that in addition to $b \cdot z < 0$ for all $z \in S$, $z \neq 0$, and $b \cdot v_i \geq 0$ for $i = 1, 2$, we also have $b \cdot \mu \geq 0$. In that case,

$$\hat{\mathbb{P}}_\mu^z(-b \cdot z \geq b \cdot W_t + b \cdot \mu t, \quad t < \infty) = 0,$$

so (4.6) and (4.5) imply (4.3). □

Proof of Proposition 4.1.5. Suppose first that $0 < \xi \leq \pi/2$. Let $z \in S^0$ and $\mu \in \mathbb{R}^2$ be given by $\mu = \|\mu\|(\cos \eta, \sin \eta) \neq 0$, where $\eta \in (0, \xi)$. Next, set

$$X_t = z + B_t + \mu t, \quad t \geq 0, \tag{4.7}$$

where B_t is a standard 2-d Brownian motion.

Now translate the origin of the coordinate axes to z and then rotate the axis in a counterclockwise direction by the angle η . By the translational and rotational invariance of Brownian motion, in these new coordinates the process X may be written as

$$\hat{X}_t = \hat{B}_t + \|\mu\|\hat{e}_1 t, \quad t \geq 0,$$

where $\hat{B}$ is a standard Brownian motion and $\hat{e}_1 = (1, 0)$. Next, in the new coordinate system denote by $\mathcal{L}_1$ the line corresponding to ∂S_1 , and by $\mathcal{L}_2$ the line corresponding to ∂S_2 . Then, in the new coordinate system, the interior of S may be expressed as

$$\mathcal{S}^0 = \{\hat{z} \in \mathbb{R}^2 : \mathcal{L}_1(\hat{z}_1) < \hat{z}_2 < \mathcal{L}_2(\hat{z}_1)\},$$

where $\mathcal{L}_i(\hat{z}_1)$ is a coordinate uniquely determined by the relation $(\hat{z}_1, \mathcal{L}_i(\hat{z}_1)) \in \mathcal{L}_i$, for $i = 1, 2$.

Hence, to complete the proof for the case $0 < \xi \leq \pi/2$, it suffices to show that

$$P(\mathcal{L}_1(\hat{X}_t^1) < \hat{X}_t^2 < \mathcal{L}_2(\hat{X}_t^1), t \geq 0) > 0.$$

First note that since $0 < \eta < \xi \leq \pi/2$, we may write

$$\mathcal{L}_1(\hat{z}_1) = -a - b\hat{z}_1 \quad \text{and} \quad \mathcal{L}_2(\hat{z}_1) = c + d\hat{z}_1,$$

for $a, b, c, d > 0$. Hence,

$$\begin{aligned} & \{\mathcal{L}_1(\hat{X}_t^1) < \hat{X}_t^2 < \mathcal{L}_2(\hat{X}_t^1), t \geq 0\} \\ &= \{-a - b\|\mu\|t - b\hat{B}_t^1 < \hat{B}_t^2 < c + d\|\mu\|t + d\hat{B}_t^1, t \geq 0\}. \end{aligned} \tag{4.8}$$

From (4.8), it now follows after some algebra that

$$\begin{aligned} & \{\hat{B}_t^1 > \max(-a/2b, -c/2d) - (\|\mu\|/2)t, t \geq 0\} \\ & \cap \{-(a/2) - (b/2)\|\mu\|t < \hat{B}_t^2 < (c/2) + (d/2)\|\mu\|t, t \geq 0\} \\ & \subseteq \{\mathcal{L}_1(\hat{X}_t^1) < \hat{X}_t^2 < \mathcal{L}_2(\hat{X}_t^1), t \geq 0\}. \end{aligned}$$

Next, by the independence of $\hat{B}^1$ and $\hat{B}^2$, we have

$$\begin{aligned} & P(\mathcal{L}_1(\hat{X}_t^1) < \hat{X}_t^2 < \mathcal{L}_2(\hat{X}_t^1), t \geq 0) \\ & \geq P(\hat{B}_t^1 > \max(-a/2b, -c/2d) - (\|\mu\|/2)t, t \geq 0) \\ & \quad \times P(-(a/2) - (b/2)\|\mu\|t < \hat{B}_t^2 < (c/2) + (d/2)\|\mu\|t, t \geq 0). \end{aligned}$$

However, since $a, b, c, d, \|\mu\| > 0$, it follows by (4.3) of Doob [14] that the two probabilities on the right-hand side above are greater than zero. This completes the proof for the case

$0 < \xi \leq \pi/2$.

Now suppose that $\pi/2 < \xi < \pi$ and $z \in S^0$ and $0 < \eta < \xi$. In this case, we show that there exists a wedge $\bar{S} \subset S$ such that $z \in \bar{S}^0$ and $P(X_t \in \bar{S}^0, t \geq 0) > 0$, which is sufficient to complete the proof. First, suppose that $0 < \eta < \pi/2$. In this case, the wedge $\bar{S}$ can be defined in the usual way by setting $\xi = \pi/2$ and placing the vertex of $\bar{S}$ at a point $\bar{z} \in S^0$ such that $\bar{z} < z$. The results above then yield the desired conclusion. Next, suppose $\pi/2 \leq \eta < \xi$. Then set

$$\bar{S} = v + \{r \geq 0, \pi - (\xi + \eta)/2 \leq \theta \leq (\xi + \eta)/2\},$$

where the vertex $v = (v_1, v_2) \in S^0$ is such that $v_1 = z_1$ and $v_2 < z_2$. The results above again yield the desired result. $\square$

Chapter 5

Submartingale problem with drift, its Markov and Feller properties

5.1 Submartingale problem with drift, its Markov and Feller properties

Before stating our main results, we still recall the main notations and definitions from Chapter 3. Let $C_S = C(\mathbb{R}_+, S)$ and, for each $t \geq 0$, let $Z(t) : C_S \rightarrow S$ denote the coordinate map $Z(t)(\omega) = \omega(t)$ for $\omega \in C_S$. Also, let $Z = \{Z(t), t \geq 0\}$ denote the coordinate mapping process on C_S . Let $\mathcal{M}_t = \sigma(Z(s), 0 \leq s \leq t)$ be the underlying natural filtration with terminal σ -algebra $\mathcal{M} = \sigma(Z(s), s \geq 0)$. For each $n \geq 1$ and domain $\Omega \subseteq \mathbb{R}^2$, we denote by $C_b^n(\Omega)$ the set of n times bounded continuously differentiable functions on Ω . We assume that the wedge S is positioned so that one side of it is the positive horizontal half line, and the angle of the wedge is ξ . We define ∂S_1 and ∂S_2 as the two sides of the wedge so that neither includes the vertex, i.e., $\partial S_1 = \{(x, 0) : x > 0\}$ and $\partial S_2 = \{r(\cos \xi, \sin \xi) : r > 0\}$. Next (see Figure 3.1), we denote by v_1 and v_2 the reflection directions on the boundaries ∂S_1 and ∂S_2 , respectively. For convenience, we assume that each v_i is normalized such that $v_i \cdot n_i = 1$, where n_i is the inward facing normal vector on ∂S_i for $i = 1, 2$. Finally, for $i = 1, 2$, we set

the directional derivative operator $D_i = v_i \cdot \nabla$, with ∇ being the gradient operator, the dot is the inner product, and denote by Δ the Laplacian operator.

Again, for our convenience, we attach the figure of the state-space here again.

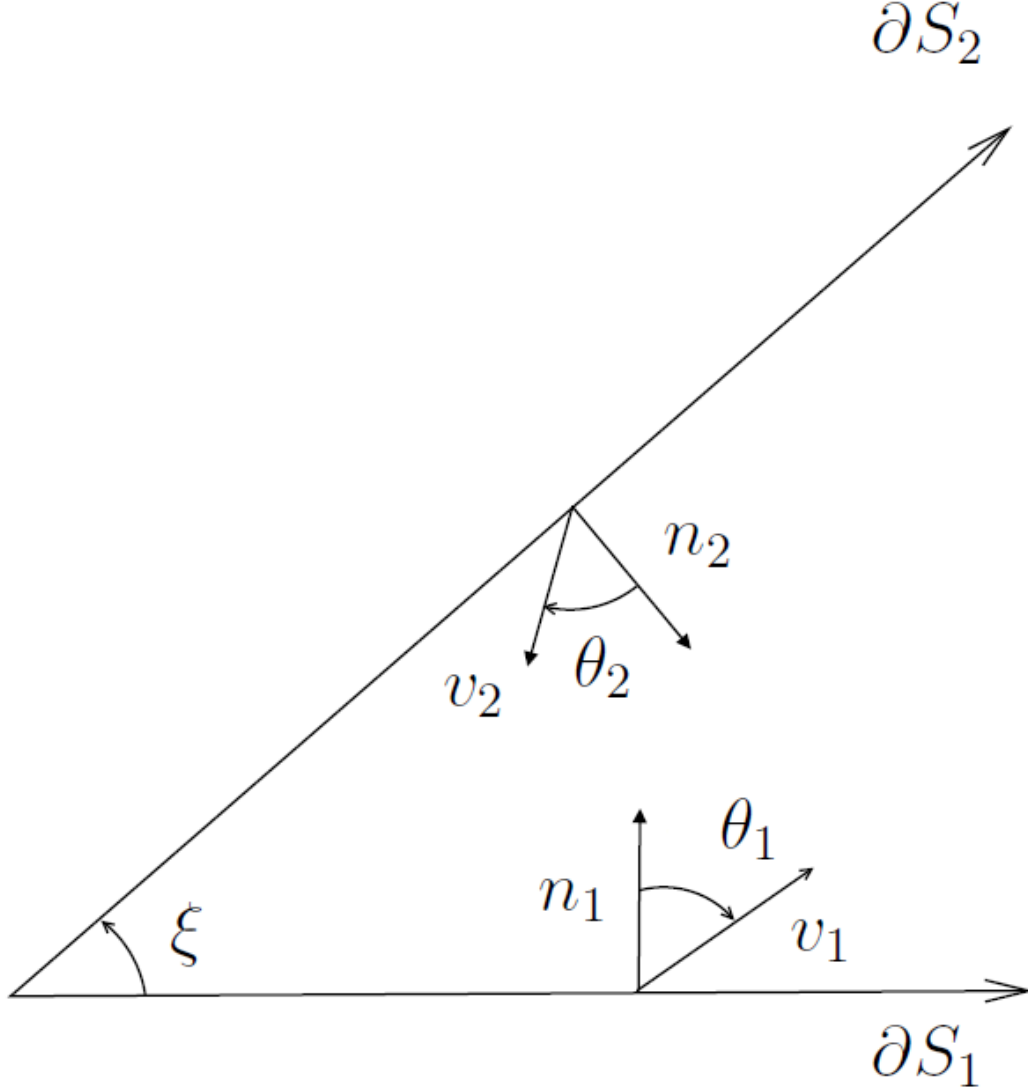


Figure 5.1: RBM in a Wedge

5.2 The problem and the main results

Definition 6 (Submartingale Problem with Drift). *A family of probability measures $\{\mathbb{P}_\mu^z, z \in S\}$ on $(C_S, \mathcal{M})$ is said to solve the submartingale problem with drift $\mu \in \mathbb{R}^2$ if for each $z \in S$,*

the following three conditions hold,

1. $\mathbb{P}_\mu^z(Z(0) = z) = 1;$

2. For each $f \in C_b^2(S)$, the process

$$\left\{ f(Z(t)) - \int_0^t \mu \cdot \nabla f(Z(s)) ds - \frac{1}{2} \int_0^t \Delta f(Z(s)) ds, t \geq 0 \right\}$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ whenever f is constant in a neighbourhood of the origin and satisfies $D_i f \geq 0$ on ∂S_i for $i = 1, 2$;

3. $\mathbb{E}_\mu^z \left[\int_0^\infty \mathbf{1}_{\{Z(t)=0\}} dt \right] = 0.$

The above definition bears a relationship to the extended Skorokhod problem (ESP) developed in [37]. We shall recall the definition of the ESP below. Let $d(\cdot)$ be a set-valued map from ∂S , the boundary of S , to the class of subsets of $\mathbb{R}^2$ satisfying the following two conditions:

(d1) for any $x \in \partial S$, the image $d(x)$ is a non-empty closed convex cone in $\mathbb{R}^2$ with the vertex being the origin;

(d2) the graph $\{(x, d(x)); x \in \partial S\}$ is closed.

For convenience, we extend the definition of $d(\cdot)$ to S by setting $d(x) = \{0\}$ for all $x \in S^\circ$, where S° is the interior of S . For a set $A \subset \mathbb{R}^2$, let $\text{co}(A)$ be the closed convex cone generated by A .

Definition 7 (Extended Skorokhod Problem (ESP)). *A pair of processes $(\phi, \eta) \in C_S \times C(\mathbb{R}_+, \mathbb{R}^2)$ is said to solve the ESP $(S, d(\cdot))$ for $\psi \in C(\mathbb{R}_+, \mathbb{R}^2)$ such that $\psi(0) \in S$ if $\phi(0) = \psi(0)$, and if for all $t \in \mathbb{R}_+$, the following properties hold,*

1. $\phi(t) = \psi(t) + \eta(t);$

2. $\phi(t) \in S;$

3. For every $s \in [0, t]$;

$$\eta(t) - \eta(s) \in co\left[\bigcup_{u \in (s, t]} d(\phi(u))\right].$$

Item 2 in the above definition is redundant since we already required that $\phi \in C_S$, but we kept that item as it appears in the original definition in [37].

Just like in [51], let

$$\alpha = \frac{\theta_1 + \theta_2}{\xi}.$$

The quantity α plays a prominent role.

Theorem 5.2.1. *If $\alpha < 2$, then for each $\mu \in \mathbb{R}^2$ there exists a unique solution $\{\mathbb{P}_\mu^z, z \in S\}$ to the submartingale problem with drift. In addition, the following statements hold:*

1. *There exists a process X defined on $(C_S, \mathcal{M}, \mathcal{M}_t)$ which, for each $z \in S$, is a 2-dimensional Brownian motion with drift μ started at z under $\mathbb{P}_\mu^z$;*
2. *Setting $Y = Z - X$, the pair (Z, Y) solves the ESP $(S, d(\cdot))$ for X , $\mathbb{P}_\mu^z$ -a.s., for each $z \in S$.*

The above theorem establishes a decomposition

$$Z = X + Y, \tag{5.1}$$

such that for all $z \in S$ under $\mathbb{P}_\mu^z$ the process X is a standard Brownian motion with drift μ started at z . In the following two theorems, we shall establish several properties of the process Y appearing in the above decomposition. In order to state the first of these two results, we need the definition of the strong p -variation of a function. Let $T > 0$ be arbitrary. We call an ordered set $(t_0, t_1, \dots, t_n)$ a partition of the interval $[0, T]$, if $0 = t_0 < t_1 < \dots < t_n = T$, for an arbitrary $n \in \mathbb{N}_+$. Let $\pi(T)$ denote the set of all partitions of the interval $[0, T]$. We

define the mesh of a partition $\rho = (t_0, \dots, t_n) \in \pi(T)$ by setting

$$\|\rho\| = \max\{t_i - t_{i-1} : i = 1, \dots, n\}.$$

Definition 8. Let $T > 0$ and $p > 0$. The strong p -variation of a function $f : \mathbb{R}_+ \mapsto \mathbb{R}^k$ on $[0, T]$ is defined by

$$V_p(f, [0, T]) = \sup \left\{ \sum_{t_i \in \rho, i \geq 1} \|f(t_i) - f(t_{i-1})\|^p : \rho \in \pi(T) \right\}.$$

Theorem 5.2.2. Suppose that $1 < \alpha < 2$. Then for each $p > \alpha$ and $z \in S$,

$$\mathbb{P}_\mu^z(V_p(Y, [0, T]) < +\infty) = 1, \quad T > 0, \quad (5.2)$$

and, for each $0 < p \leq \alpha$,

$$\mathbb{P}_\mu^0(V_p(Y, [0, T]) < +\infty) = 0, \quad T > 0. \quad (5.3)$$

A 2-dimensional continuous process U defined on $(C_s, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ is said to be of *zero energy* if for each $T > 0$ and each sequence of partitions $(\rho^m) \subset \pi(T)$ such that $\|\rho^m\| \rightarrow 0$ as $m \rightarrow \infty$ we have

$$\sum_{i=1}^{n(m)} \|U(t_i^m) - U(t_{i-1}^m)\|^2 \xrightarrow{\mathbb{P}_\mu^z} 0 \text{ as } m \rightarrow \infty,$$

where $\rho^m = (t_0^m, \dots, t_{n(m)}^m)$. A process D on $(C_s, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ is said to be a *Dirichlet process* if it has a decomposition $D = M + U$, where M is a local martingale on the same probability space, and U is a continuous zero-energy process with $U(0) = 0$.

Theorem 5.2.3. Let $1 < \alpha < 2$, and $z \in S$ arbitrary. Then the process Y in decomposition (5.1) is a zero-energy process, and Z is a Dirichlet process.

Theorem 5.2.4. If $1 \leq \alpha < 2$, then Z is not a semimartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$, for any

$z \in S$.

Theorem 5.2.5. *If $\alpha \geq 2$, then for any $\mu \in \mathbb{R}^2$ there is no solution to the submartingale problem with drift.*

Let $\{\mathbb{P}_\mu^z, z \in S\}$ be the solution to the submartingale problem for some $\mu \in \mathbb{R}^2$. We say that $\{\mathbb{P}_\mu^z, z \in S\}$ possesses the strong Markov property if for each stopping time τ and $z \in S$, and each bounded $\mathcal{M}$ -measurable function $h : C_S \rightarrow \mathbb{R}$ we have that

$$\mathbb{E}_\mu^z[\mathbb{1}_{\{\tau < \infty\}} f(\omega(\cdot + \tau)) | \mathcal{M}_\tau] = \mathbb{1}_{\{\tau < \infty\}} \mathbb{E}_\mu^{\omega(\tau)}[f(w(\cdot))], \quad \mathbb{P}_\mu^z\text{-a.s.} \quad (5.4)$$

Theorem 5.2.6. *If $\alpha < 2$, then for each $\mu \in \mathbb{R}^2$ the solution to the submartingale problem with drift has the strong Markov property.*

The last subject of this subsection is the Feller property of $\{\mathbb{P}_\mu^z, z \in S\}$. There are various, slightly differing definitions of the Feller property available in the literature. For clarity, we list below three definitions.

1. We say that $\{\mathbb{P}_\mu^z, z \in S\}$ has the Feller property if for any $\{z_n, n \geq 1\} \subset S$ converging to $z \in S$, $\mathbb{P}_\mu^{z_n} \Rightarrow \mathbb{P}_\mu^z$ as $n \rightarrow \infty$ (see Varadhan and Williams [51]).
2. Let $\hat{C}(S)$ be the set of continuous functions on S vanishing at infinity. We say that $\{\mathbb{P}_\mu^z, z \in S\}$ has the $\hat{C}(S)$ -Feller property if for any $f \in \hat{C}(S)$, and $t \geq 0$, the function $z \mapsto \mathbb{E}_\mu^z[f(Z_t)]$ is also in $\hat{C}(S)$.
3. We say that $\{\mathbb{P}_\mu^z, z \in S\}$ has the $C_b(S)$ -Feller property if for any $f \in C_b(S)$, and $t \geq 0$, the function $z \mapsto \mathbb{E}_\mu^z[f(Z_t)]$ is also in $C_b(S)$.

Remark 5.2.7. The Feller property obviously implies the $C_b(S)$ -Feller property. The $\hat{C}(S)$ -Feller property implies the $C_b(S)$ -Feller, but the converse is not true (cf. Theorems 1.9 and

1.10 in [2]).

Theorem 5.2.8. *If $\alpha < 2$, then the solution to the submartingale problem for each $\mu \in \mathbb{R}^2$ has the Feller property.*

Theorem 5.2.9. *If $\alpha < 2$, then the solution to the submartingale problem for each $\mu \in \mathbb{R}^2$ has the $\hat{C}(S)$ -Feller property.*

We note that for the $\mu = 0$ case, the Feller property is known ([51], Theorem 3.13). However, the $\hat{C}(S)$ -Feller property is new even in the $\mu = 0$ case.

5.3 Proof of Theorems 5.2.1, 5.2.2, 5.2.3, 5.2.4, and 5.2.5

We provide an extension of Theorems 2.4 and 2.8 in [31] to all $\alpha < 2$. For $z \in \partial S_i, i = 1, 2$, let $d(z) = \{\lambda v_i, \lambda \geq 0\}$, and set $d(0) = \mathbb{R}^2$. It is known that in the case of $\mu = 0$ the submartingale problem has a unique solution whenever $\alpha < 2$ (see [51]). In accordance with our notation, that solution will be denoted by $\{\mathbb{P}_0^z, z \in S\}$. We then have the following.

Proposition 5.3.1. *Let $(C_S, \mathcal{M}, \mathcal{M}_t)$ and Z be defined as in Section 5.2. Then, if $\alpha < 2$,*

1. *There exists a process X defined on $(C_S, \mathcal{M}, \mathcal{M}_t)$ which, for each $z \in S$, is a 2-dimensional Brownian motion started at z under $\mathbb{P}_0^z$;*
2. *Setting $Y = Z - X$, the pair (Z, Y) solves the ESP $(S, d(\cdot))$ for X , $\mathbb{P}_0^z$ -a.s..*

Proof. Let $\alpha < 2$. Then, Condition 1 is immediate from Theorem 2.4 in [31]. It remains to show that for each $z \in S$, $\mathbb{P}_0^z$ -a.s., Z and $Y = Z - X$ together solve the ESP (see Definition 7 as above) for X with d as defined immediately preceding the statement of the proposition. For $\alpha \in (1, 2)$, this follows by Theorem 2.8 in [31]. We now claim that if $\alpha \leq 1$, (Z, Y) also

solves the ESP $(S, d(\cdot))$ for X , $\mathbb{P}^z$ -a.s. For any two real numbers $0 < s < t$, if (s, t) belongs to a single excursion from the origin then by a similar proof to the one in part 2 of Theorem 4.2 in [31], one can conclude that item 3 in the definition of the ESP holds. If (s, t) doesn't belong to one excursion, then item 3 is obviously satisfied by $d(0) = \mathbb{R}^2$. $\square$

We are now ready to prove the existence of a solution to the submartingale problem with drift, and some of the properties of the solution we create.

Proposition 5.3.2. *If $\alpha < 2$, then for each $\mu \in \mathbb{R}^2$ the submartingale problem with drift has a solution $\{\mathbb{P}_\mu^z, z \in S\}$, which satisfies the following properties. With X and Y defined in Proposition 5.3.1, for every $z \in S$ the following hold:*

- Under $\mathbb{P}_\mu^z$ the process X is a standard Brownian motion with drift μ started at z ;
- The pair (Z, Y) solves the ESP $(S, d(\cdot))$ for X , $\mathbb{P}_\mu^z$ -s.s.

Proof. Let $\alpha < 2$ and note that by Proposition 5.3.1 there exists a process X defined on $(C_S, \mathcal{M}, \mathcal{M}_t)$ which, for each $z \in S$, is a 2-dimensional Brownian motion started at z under $\mathbb{P}_0^z$. Now let $T \geq 0$ and for each $z \in S$, let $\mathbb{P}_{\mu, T}^z$ be a probability measure on $(C_S, \mathcal{M}, \mathcal{M}_t)$ equivalent (mutually absolutely continuous) to $\mathbb{P}_0^z$ such that under $\mathbb{P}_{\mu, T}^z$, X is a standard Brownian motion with drift μ up to time T , started at z . In other words, $\{X(t) - \mu t, t \leq T\}$ is a standard (driftless) $\mathbb{P}_{\mu, T}^z$ -Brownian motion started at z . The measure $\mathbb{P}_{\mu, T}^z$ is defined by

$$\frac{d\mathbb{P}_{\mu, T}^z}{d\mathbb{P}_0^z} = \zeta(T), \quad (5.5)$$

where $\zeta(T) = \exp\{\mu \cdot (X(T) - X(0)) - \frac{1}{2}\|\mu\|^2 T\}$.

One can easily show that the family of probability measures $\{\mathbb{P}_{\mu, T}^z, T \in [0, \infty)\}$ is consistent. That is, if $S < T$, then $\mathbb{P}_{\mu, T}^z(A) = \mathbb{P}_{\mu, S}^z(A)$, whenever $A \in \mathcal{M}_S$. From [34], Theorem 4.2 (page 143), it follows that there exists a single probability measure $\mathbb{P}_\mu^z$ such that $\mathbb{P}_\mu^z(A) = \mathbb{P}_{\mu, T}^z(A)$ whenever $A \in \mathcal{M}_T$. Since $\{X(t) - X(0) - \mu t, t \leq T\}$ is a $\mathbb{P}_{\mu, T}^z$ -Brownian motion started at zero for every $T \in [0, \infty)$, it follows that $\{X(t) - X(0) - \mu t, t < \infty\}$ is also a $\mathbb{P}_\mu^z$ -Brownian

motion started at zero. Also, (Z, Y) solves the ESP $(S, d(\cdot))$ for X under $\mathbb{P}_\mu^z$ -a.s., because by Proposition 5.3.1 it is true under $\mathbb{P}_0^z$, and the measures $\mathbb{P}_0^z$ and $\mathbb{P}_\mu^z$ constrained to $\mathcal{M}_T$ are mutually absolutely continuous for every $T \in [0, \infty)$. Now let

$$W(t) = X(t) - X(0) - \mu t, \quad t \in [0, \infty),$$

and consider the definition of a weak solution to an SDER (see Definition 2.4 of Kang and Ramanan [27]). Clearly, the triplet $(C_S, \mathcal{M}, \mathcal{M}_t)$, $\mathbb{P}_\mu^z$, (Z, W) is a weak solution to the SDER with initial condition z associated with $(S, d(\cdot))$, $b(\cdot)$ and $\sigma(\cdot)$, where $b(x) = \mu$ and $\sigma(x) = \text{Id}_{2 \times 2}$. We note that the “closed graph condition” (see Kang and Ramanan [27], page 5) is satisfied. From Theorem 2 in [27], it now follows that $\{\mathbb{P}_\mu^z, z \in S\}$ solves the submartingale problem with drift μ .

□

We shall use the Lemmas 5.3.3, 5.3.4, 5.3.5, 5.3.6, and 5.3.7 for the proof of both the uniqueness part of Theorem 5.2.1, and for the proof of Theorem 5.2.5. In these lemmas, α may be an arbitrary real number. On the other hand, in these lemmas we start with a probability measure P_μ^z that satisfies conditions 1, 2, and 3 of Definition 6. This may be surprising since Theorem 5.2.5 states that such probability measure does not exist for $\alpha \geq 2$. However, for such α 's we use these lemmas to derive a contradiction, thereby proving Theorem 5.2.5.

Lemma 5.3.3. *Suppose that $\{\mathbb{P}_\mu^z, z \in S\}$ is a solution to the submartingale problem with drift $\mu \in \mathbb{R}^2$. Then, for all $z \in S$,*

$$\mathbb{E}_\mu^z \left[\int_0^\infty \mathbf{1}_{\{Z(t) \in \partial S\}} dt \right] = 0. \quad (5.6)$$

Proof. Let $z \in S$ be arbitrary. In this proof, we shall use the Doob-Meyer decomposition for submartingales, which requires that the probability space is augmented. For this reason we denote by $(C_S, \mathcal{F}^z, (\mathcal{F}_t^z))$ the augmentation of the space $(C_S, \mathcal{M}, (\mathcal{M}_t))$ under $\mathbb{P}_\mu^z$. For

some technical details on the augmentation of probability spaces see Remark A.1.2 in the Appendix. Condition 3 of the submartingale problem gives

$$\mathbb{E}_\mu^z \left[\int_0^\infty \mathbb{1}_{\{Z(t)=0\}} dt \right] = 0,$$

for each $z \in S$, thus in order to complete the proof it suffices to prove that

$$\mathbb{E}_\mu^z \left[\int_0^\infty \mathbb{1}_{\{Z(t) \in \partial S_i\}} dt \right] = 0, \text{ for } i = 1, 2. \quad (5.7)$$

We prove this result for $i = 1$; the result then follows for $i = 2$ by symmetry.

For each $\varepsilon > 0$, define $S^\varepsilon \subset S$ by $S^\varepsilon = S + (\varepsilon, 0)$, i.e., a wedge with vertex at $(\varepsilon, 0)$ and edges $\partial S_1^\varepsilon = \{(x, 0) \in \mathbb{R}^2, x > \varepsilon\}$ and $\partial S_2^\varepsilon = \{(\varepsilon, 0) + \lambda(\cos \xi, \sin \xi), \lambda > 0\}$ (recall that ξ is the angle of the wedge S).

Next we shall recursively define the $(\mathcal{F}_t^z)$ stopping times $\bar{\sigma}_k^{\varepsilon, T}, \bar{\tau}_k^{\varepsilon, T}$ for $k \geq 1$, for every $T > 0$. We define

$$\bar{\sigma}_1^{\varepsilon, T} = \inf \left\{ t \geq 0 : Z(t) \in S^\varepsilon \right\} \wedge T,$$

and

$$\bar{\tau}_k^{\varepsilon, T} = \inf \left\{ t \geq \bar{\sigma}_k^{\varepsilon, T} : Z_t \in \partial S_2^{2\varepsilon/3} \right\} \wedge T, \quad k \geq 1,$$

$$\bar{\sigma}_k^{\varepsilon, T} = \inf \left\{ t \geq \bar{\tau}_{k-1}^{\varepsilon, T} : Z_t \in \partial S_2^\varepsilon \right\} \wedge T, \quad k \geq 2;$$

Let $Z_t = (Z_1(t), Z_2(t))$. Let $C > z_2$ be an arbitrary constant. We define the $(\mathcal{F}_t^z)$ stopping time

$$T_C = \inf \left\{ t \geq 0 : Z_t^{(2)} \geq C \right\},$$

and in order to simplify the notation, we also introduce the stopping times $\bar{\sigma}_k = \bar{\sigma}_k^{\varepsilon, T} \wedge T_C$ and $\bar{\tau}_k = \bar{\tau}_k^{\varepsilon, T} \wedge T_C$. Notice that for all $t \leq T$, $t \in [\bar{\sigma}_k, \bar{\tau}_k]$ implies that $Z_t \in S^{2\varepsilon/3}$ and $Z_2(t) \leq C$.

Let $f_{\varepsilon,C} \in C_b^2(S)$ such that

$$f_{\varepsilon,C}(x, y) = \begin{cases} 0, & \text{if } (x, y) \in S \setminus S^{\varepsilon/3}, \\ y, & \text{if } (x, y) \in S^{2\varepsilon/3}, y \leq C. \end{cases} \quad (5.8)$$

In addition we require that $f_{\varepsilon,C}(x, 0) = 0$ for all $x \geq 0$, and $D_2 f_{\varepsilon,C} \geq 0$ on ∂S_2 . It follows from (5.8) that $D_1 f_{\varepsilon,C} = 0$ on ∂S_1 . We show in Lemma A.1.1 in the Appendix that such a function indeed exists. By the definition of the submartingale problem

$$V_1 = \left\{ V_1(t) = f_{\varepsilon,C}(Z_t) - \int_0^t \left(\mu \cdot \nabla f_{\varepsilon,C}(Z_s) - \frac{1}{2} \Delta f_{\varepsilon,C}(Z_s) \right) ds; \ t \geq 0 \right\} \quad (5.9)$$

is a regular submartingale under $\mathbb{P}_\mu^z$ on $(\mathcal{F}_t^z)$, thus by Theorem 1.4.14 in [28] it has a unique Doob-Meyer decomposition

$$V_1(t) = M(t) + A(t) \quad (5.10)$$

where M is a continuous martingale and A is a continuous increasing process. For the definition of regular submartingales see Definition 1.4.12 in [28]. For an arbitrary $k \geq 1$ we have $f_{\varepsilon,C}(Z_t) = Z_2(t)$, $\mu \cdot \nabla f_{\varepsilon,C}(Z_t) = \mu_2$, and $\Delta f_{\varepsilon,C}(Z_t) = 0$ whenever $t \in [\bar{\sigma}_k, \bar{\tau}_k]$, hence by (5.10) and (5.9)

$$Z_2(t) = Z_2(\bar{\sigma}_k) + M(t) - M(\bar{\sigma}_k) + A(t) - A(\bar{\sigma}_k) + \mu_2(t - \bar{\sigma}_k) \quad (5.11)$$

for $t \in [\bar{\sigma}_k, \bar{\tau}_k]$. Next, we are going to establish the following two properties. The first is that

$$\int_{\bar{\sigma}_k}^{\bar{\tau}_k} 1_{\{Z_2(t) > 0\}} dA(t) = 0, \quad \mathbb{P}_\mu^z\text{-a.s.}, \quad (5.12)$$

and the second is that

$$\int_{\bar{\sigma}_k}^{\bar{\tau}_k} d(\langle M \rangle_t - t) = 0, \quad \mathbb{P}_\mu^z\text{-a.s.} \quad (5.13)$$

We start with proving (5.12). For any $\delta > 0$ and $k \geq 1$ we define a sequence of $(\mathcal{F}_t^z)$ stopping

times

$$\theta_1^\delta = \inf\{t \geq \bar{\sigma}_k : Z_z(t) \geq \delta\} \wedge \bar{\tau}_k, \quad \vartheta_1^\delta = \inf\left\{t \geq \theta_1^\delta : Z_2(t) = \frac{\delta}{2}\right\} \wedge \bar{\tau}_k,$$

$$\theta_n^\delta = \inf\{t \geq \vartheta_{n-1}^\delta : Z_2(t) = \delta\} \wedge \bar{\tau}_k, \quad n \geq 2,$$

$$\vartheta_n^\delta = \inf\left\{t \geq \theta_n^\delta : Z_2(t) = \frac{\delta}{2}\right\} \wedge \bar{\tau}_k, \quad n \geq 2.$$

Notice that $[\theta_n^\delta, \vartheta_n^\delta]$ is a sub-interval of $[\bar{\sigma}_k, \bar{\tau}_k]$ such that for $t \in [\theta_n^\delta, \vartheta_n^\delta]$ we have $Z_2(t) \geq \delta/2$. All these stopping times are finite because by definition $\bar{\tau}_k \leq T$. Let $g_1 \in C_b^2(\mathbb{R})$ be an arbitrary function such that $g_1'(0) = 0$, and $g_1(x) = x$ whenever $x \geq \delta/2$. Relation $g_1'(0) = 0$ implies that

$$V_2 = \left\{ V_2(t) = g_1(f_{\varepsilon, C}(Z_t)) - \int_0^t \left(\mu \cdot \nabla(g_1 \circ f_{\varepsilon, C})(Z_s) - \frac{1}{2} \Delta(g_1 \circ f_{\varepsilon, C})(Z_s) \right) ds, \quad t \geq 0 \right\}$$

is a martingale with respect to the filtration $(\mathcal{F}_t^z)$ under $\mathbb{P}_\mu^z$. Therefore, $\{V_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta), t \geq 0\}$ is also a martingale with respect to the filtration $\{\mathcal{F}_{(t \vee \theta_n^\delta) \wedge \vartheta_n^\delta}^z, t \geq 0\}$ under $\mathbb{P}_\mu^z$. For all $s \in [\theta_n^\delta, \tau_n^\delta]$ we have $g_1(f_{\varepsilon, C}(Z(s))) = Z_2(s)$, $\mu \cdot \nabla(g_1 \circ f_{\varepsilon, C})(Z(s)) = \mu_2$ and $\Delta(g_1 \circ f_{\varepsilon, C})(Z(s)) = 0$, hence for all $t \geq 0$

$$V_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) = V_2(\theta_n^\delta) - Z_2(\theta_n^\delta) + Z_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) - \mu_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta - \theta_n^\delta)$$

thus $\{Z_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) - \mu_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta), t \geq 0\}$ is also a martingale with respect to the filtration $\{\mathcal{F}_{(t \vee \theta_n^\delta) \wedge \vartheta_n^\delta}^z, t \geq 0\}$ under $\mathbb{P}_\mu^z$. On the other hand, from (5.11) follows that

$$Z_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) - \mu_2((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) =$$

$$Z_2(\theta_n^\delta) - \mu_2\theta_n^\delta + M((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) - M(\theta_n^\delta) + A((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta) - A(\theta_n^\delta),$$

for all $t \geq 0$. However, the left-hand side in the above identity is a martingale with respect to

the filtration $\left\{ \mathcal{F}_{(t \vee \theta_n^\delta) \wedge \vartheta_n^\delta}^z, t \geq 0 \right\}$ under $\mathbb{P}_\mu^z$, and so is $M((t \vee \theta_n^\delta) \wedge \vartheta_n^\delta)$ on the right-hand side ($t \geq 0$). Therefore, A must be constant on $[\theta_n^\delta, \tau_n^\delta]$, $\mathbb{P}_\mu^z$ -a.s. This holds for all $n \geq 1$, hence

$$\int_{\hat{\sigma}_k}^{\hat{\tau}_k} \sum_{n=1}^{\infty} 1_{[\theta_n^\delta, \vartheta_n^\delta]}(t) dA(t) = 0, \quad \mathbb{P}_\mu^z\text{-a.s.} \quad (5.14)$$

If $t \in [\bar{\sigma}_k, \bar{\tau}_k]$ and $Z_2(t) > \delta$, then $Z_2(t) \in [\theta_n^\delta, \vartheta_n^\delta]$ for some $n \geq 1$, hence by (5.14)

$$\int_{\hat{\sigma}_k}^{\hat{\tau}_k} 1_{\{Z_2(t) > \delta\}}(t) dA(t) = 0, \quad \mathbb{P}_\mu^z\text{-a.s.},$$

and (5.12) follows.

Next, we are going to show (5.13). Let $g_2 \in C_b^2(\mathbb{R})$ arbitrary such that $g_2(x) = x^2$ whenever $|x| \leq C$. Then

$$V_3 = \left\{ V_3(t) = g_2(f_{\varepsilon, C}(Z_t)) - \int_0^t \left(\mu \cdot \nabla(g_2 \circ f_{\varepsilon, C})(Z_s) - \frac{1}{2} \Delta(g_2 \circ f_{\varepsilon, C})(Z_s) \right) ds, t \geq 0 \right\} \quad (5.15)$$

is a martingale under $\mathbb{P}_\mu^z$ with respect to the filtration $(\mathcal{F}_t^z)$, and for $t \in [\bar{\sigma}_k, \bar{\tau}_k]$ we have $g_2(f_{\varepsilon, C}(Z_t)) = (Z_2(t))^2$, $\mu \cdot \nabla(g_2 \circ f_{\varepsilon, C})(Z_t) = 2\mu_2 Z_2(t)$ and $\Delta(g_2 \circ f_{\varepsilon, C})(Z_t) = 2$, hence by Ito's rule applied to $g_2(f_{\varepsilon, C}(Z_t))$ and by (5.11)

$$Z_2^2(t) = Z_2^2(\bar{\sigma}_k) + \int_{\bar{\sigma}_k}^t 2Z_2(s) dM(s) + \int_{\bar{\sigma}_k}^t 2\mu_2 Z_2(s) ds + \langle M \rangle_t - \langle M \rangle_{\bar{\sigma}_k},$$

for $t \in [\bar{\sigma}_k, \bar{\tau}_k]$. We note that the $\int_{\bar{\sigma}_k}^t 2Z_2(s) dA(s)$ term vanished because of (5.12). From this and from (5.15) follows that

$$V_3(t) = V_3(\bar{\sigma}_k) + \int_{\bar{\sigma}_k}^t 2Z_2(s) dM(s) + \int_{\bar{\sigma}_k}^t d(\langle M \rangle_s - s),$$

for $t \in [\bar{\sigma}_k, \bar{\tau}_k]$. The process $\{V_3((t \vee \bar{\sigma}_k) \wedge \bar{\tau}_k), t \geq 0\}$ is a martingale with respect to the filtration $\left\{ \mathcal{F}_{(t \vee \theta_n^\delta) \wedge \vartheta_n^\delta}^z, t \geq 0 \right\}$ under $\mathbb{P}_\mu^z$, and can be written by substituting $(t \vee \bar{\sigma}_k) \wedge \bar{\tau}_k$ for t

in the above identity as

$$V_3(t \vee \bar{\sigma}_k) \wedge \bar{\tau}_k = V_3(\bar{\sigma}_k) + \int_{\bar{\sigma}_k}^{t \vee \bar{\sigma}_k) \wedge \bar{\tau}_k} 2Z_2(s) dM(s) + \int_{\bar{\sigma}_k}^{t \vee \bar{\sigma}_k) \wedge \bar{\tau}_k} d(\langle M \rangle_s - s),$$

for all $t \geq 0$. Since the left-hand side is a martingale with respect to the filtration $\left\{ \mathcal{F}_{(t \vee \theta_n^\delta) \wedge \vartheta_n^\delta}^z, t \geq 0 \right\}$ under $\mathbb{P}_\mu^z$, (5.13) follows. The by (5.11), Z_2 is a 1-dimensional Brownian motion with drift μ_2 reflected at zero in $[\bar{\sigma}_k, \bar{\tau}_k]$. Therefore,

$$\int_{\bar{\sigma}_k}^{\bar{\tau}_k} 1_{\{Z_2(t)=0\}} dt = 0, \quad \mathbb{P}_\mu^z\text{-a.s.}$$

This holds for every $k \geq 1$, hence we also have

$$\sum_{k=1}^{\infty} \int_{\bar{\sigma}_k}^{\bar{\tau}_k} 1_{\{Z_2(t)=0\}} dt = 0, \quad \mathbb{P}_\mu^z\text{-a.s.},$$

and from this

$$\int_0^{T \wedge T_C} 1_{\{Z_1(t) \geq \varepsilon, Z_2(t)=0\}} dt = 0, \quad \mathbb{P}_\mu^z\text{-a.s.}$$

The last identity follows because $t \leq T \wedge T_C$, $Z_1(t) \geq \varepsilon$ and $Z_2(t) = 0$ implies that $t \in [\bar{\sigma}_k, \bar{\tau}_k]$ for some $k \geq 1$. The statement of the Lemma now follows by $T, C \uparrow \infty$ and $\varepsilon \downarrow 0$.

□

Let $\{\mathbb{P}_\mu^z, z \in S\}$ be a solution to the submartingale problem with a drift μ . Next we shall create a process X which is a Brownian motion with drift μ started at z on $(C_S, \mathcal{M}, (\mathcal{M}_t), \mathbb{P}_\mu^z)$, for every $z \in S$. We already know that such a process exists for the solution that we created in Proposition 5.3.2. However, for proving the uniqueness of the solution, we need to show the existence of such process X for every solution of the submartingale problem. Such construction has been carried out in [31] and in [27] for the case of zero drift. The generalization to the case of non-zero drift requires only a few obvious changes to the proofs in the case of zero drift, so here we shall only state the results (Lemmas 5.3.4 and 5.3.5) without proofs.

For each $\delta > 0$, let $S_\delta \subset S$ be the closed set defined in the complex plane by $S_\delta = S + \delta e^{i\xi/2}$. So S_δ is a wedge with vertex at $\delta (\cos (\xi/2), \sin (\xi/2))$, such that it is included in S and has edges parallel with the respective edges of S .

Set $\tau_0^\delta = 0$, and, for each $k \geq 1$, recursively define

$$\sigma_k^\delta = \inf\{t \geq \tau_{k-1}^\delta : Z(t) \in S_{2\delta}\} \text{ and } \tau_k^\delta = \inf\{t \geq \sigma_k^\delta : Z(t) \in S \setminus S_\delta\}.$$

By Problem 1.2.7 in Karatzas and Shreve [28], σ_k^δ and τ_k^δ are stopping times relative to $\{\mathcal{M}_t, t \geq 0\}$ for every $k \geq 1$.

For each $k \geq 1$ and $\delta > 0$, define the process $\{W_{(k)}^\delta(t), t \geq 0\}$ by setting

$$W_{(k)}^\delta(t) = Z(t \wedge \tau_k^\delta) - Z(t \wedge \sigma_k^\delta) - (t \wedge \tau_k^\delta - t \wedge \sigma_k^\delta)\mu, \quad t \geq 0,$$

and then define the process $\{W^\delta(t), t \geq 0\}$ by setting

$$W^\delta(t) = \sum_{k=1}^{\infty} W_{(k)}^\delta(t), \quad t \geq 0.$$

Lemma 5.3.4. *For every $\delta > 0$ and $z \in S$ the process W^δ is a square-integrable martingale on $(C_S, \mathcal{M}, (\mathcal{M}_t), \mathbb{P}_\mu^z)$.*

Lemma 5.3.5. *There exists a process W on $(C_S, \mathcal{M}, \mathcal{M}_t)$ such that for every $z \in S$ it is a standard 2-dimensional Brownian motion under $\mathbb{P}_\mu^z$ starting at zero, and for every fixed $T > 0$ we have*

$$E_\mu^z [\|W(T) - W^\delta(T)\|^2] \rightarrow 0, \quad \text{as } \delta \rightarrow 0. \quad (5.16)$$

Next, we shall define the process X by

$$X(t, \omega) = \omega(0) + W(t, \omega) + \mu t, \quad t \geq 0. \quad (5.17)$$

We define the process Y by

$$Y(t) = Z(t) - X(t), \quad t \geq 0. \quad (5.18)$$

We shall say that a function $f : \mathbb{R}_+ \mapsto \mathbb{R}^2$ is *flat* on an interval $[s, t] \subset \mathbb{R}_+$, if for every $u \in [s, t]$ we have $f(u) = f(s)$.

Lemma 5.3.6. *Let $\{\mathbb{P}_\mu^z; z \in S\}$ be an arbitrary solution of the submartingale problem, and let X and Y be the processes defined above. Then the following two statements hold for every $z \in S$:*

1. *Under $\mathbb{P}_\mu^z$ the process X is a standard 2-dimensional Brownian motion on $(C_S, \mathcal{M}, \mathcal{M}_t)$ with drift μ , started at z ;*
2. *for every $n \in \mathbb{N}_+$, and $\delta > 0$, the sample paths of Y are flat on $[\sigma_n^\delta, \tau_n^\delta]$, $\mathbb{P}_\mu^z$ -a.s.*

Proof. The first statement follows from Lemma 5.3.5 and property 1 in the definition of the Submartingale Problem. Next, we shall prove the second statement. By the definition of w^δ , the sample paths of

$$\{Z_t - w^\delta(t) - \mu t, \quad t \geq 0\} \text{ are flat on } [\sigma_n^\delta, \tau_n^\delta], \quad (5.19)$$

for each $\delta > 0$, $n \geq 1$. On the other hand, for every $\delta > 0$, $n \geq 1$ there exists $k \geq 1$ (depending on the sample path) such that $[\sigma_n^\delta, \tau_n^\delta] \subset [\sigma_k^{\delta/2}, \tau_k^{\delta/2}]$. This implies that the sample paths of $\{Z_t - w^{\delta/2}(t) - \mu t, \quad t \geq 0\}$ are also flat on $[\sigma_n^\delta, \tau_n^\delta]$. Iterating this we get that for every $m \geq 1$ the sample paths of $\{Z_t - w^{\delta/2^m}(t) - \mu t, \quad t \geq 0\}$ are also flat on $[\sigma_n^\delta, \tau_n^\delta]$. Comparing this with (5.19) we conclude that the sample paths of $w^\delta - w^{\delta/2^m}$ are also flat on $[\sigma_n^\delta, \tau_n^\delta]$, thus for every $t \geq 0$

$$\int_0^t 1_{[\sigma_n^\delta, \tau_n^\delta]}(s) dw^{\delta/2^m}(s) = \int_0^t 1_{[\sigma_n^\delta, \tau_n^\delta]}(s) dw^\delta(s).$$

Taking limit as $m \rightarrow \infty$ and using (5.16) we get that

$$\int_0^t 1_{[\sigma_n^\delta, \tau_n^\delta]}(s) dw(s) = \int_0^t 1_{[\sigma_n^\delta, \tau_n^\delta]}(s) dw^\delta(s),$$

$\mathbb{P}_\mu^z$ -a.s. This identity and (5.19) imply that the sample paths of $\{Z_t - w(t) - \mu t, t \geq 0\}$ are flat on $[\sigma_n^\delta, \tau_n^\delta] \cap [0, t]$, $\mathbb{P}_\mu^z$ -a.s. Since $t \geq 0$ was arbitrary, this and (5.17) imply what we wanted to prove. $\square$

Lemma 5.3.7. *Suppose that Q_1 and Q_2 are mutually absolutely continuous probability measures on $\mathcal{M}$, both satisfying properties 1,2, and 3 of Definition 6 with $\mathbb{P}_\mu^z$ replaced by Q_i ($i = 1, 2$). Then there exist probability measures $\tilde{Q}_i$ on $\mathcal{M}$ for $i = 1, 2$, such that conditions 1,2, and 3 of Definition 6 are satisfied with P_μ^z replaced by $\tilde{Q}_i$ and μ replaced by the zero vector. Furthermore, for every $T \geq 0$ there exist probability measures $\tilde{Q}_1^T$ and $\tilde{Q}_2^T$ on $\mathcal{M}$ such that for all $T \geq 0$, $A \in \mathcal{M}_T$, and $i = 1, 2$ we have $\tilde{Q}_i^T(A) = \tilde{Q}_i(A)$, $\tilde{Q}_i^T$ and Q_i are mutually absolutely continuous, and*

$$\frac{d\tilde{Q}_1^T}{dQ_1} = \frac{d\tilde{Q}_2^T}{dQ_2}. \quad (5.20)$$

Proof. Let Q_1 and Q_2 as above, and let (X^i, Y^i) be as in Lemma 5.3.6 defined under Q_i . Since X^i is defined by $L^2(Q_i)$ convergence, this implies that $(X^1, Y^1) = (X^2, Y^2)$, which we shall from here on denote by (X, Y) . In this proof we shall use the Doob-Meyer decomposition which requires that the probability space satisfies the “usual conditions”, and for this purpose, we have to augment the probability space $(C_S, \mathcal{M}, (\mathcal{M}_t), Q_i)$, $i = 1, 2$; let this augmentation be $(C_S, \mathcal{F}, (\mathcal{F}_t), Q_i)$. The measures Q_1 and Q_2 are mutually absolutely continuous, hence the filtration $(\mathcal{F}_t)$ and the sigma field $\mathcal{F}$ do not depend on $i = 1, 2$. For technical details concerning the augmentation of a probability space please see Remark A.1.2 in the Appendix. In this proof, all processes live on the augmented space $(C_S, \mathcal{F}, (\mathcal{F}_t))$, unless specified otherwise.

Next, note that for each $T > 0$ and $\delta > 0$ and $n \geq 1$,

$$\left\{ Z(\tau_n^\delta \wedge t) - Z(\sigma_n^\delta \wedge t), \ t \in [0, T] \right\}$$

is a semimartingale under both Q_1 and Q_2 with respect to the filtration $(\mathcal{F}_t)$. Indeed, it can be written by Lemma 5.3.6 and by (5.18) as

$$Z(\tau_n^\delta \wedge t) - Z(\sigma_n^\delta \wedge t) = X(\tau_n^\delta \wedge t) - X(\sigma_n^\delta \wedge t), \quad t \in [0, T].$$

Now let $f \in C_b^2(S)$ such that $D_i f \geq 0$ on ∂S_i for $i = 1, 2$, and f is constant in a neighborhood of the origin. Then, by Itô's rule we have that for $t \in [0, T]$,

$$f(Z(t \wedge \tau_n^\delta)) = f(Z(t \wedge \sigma_n^\delta)) + \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \nabla f(Z(s)) dX(s) + \frac{1}{2} \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \Delta f(Z(s)) ds, \quad (5.21)$$

Q_i -a.s., $i = 1, 2$. On the other hand, by condition 2 of Definition 6 and by Theorems I.4.10 and I.4.14 in [28], we have for $i = 1, 2$, the unique Doob-Meyer decomposition

$$f(Z(t)) = f(z) + \int_0^t \nabla f(Z(s)) \cdot \mu ds + \frac{1}{2} \int_0^t \Delta f(Z(s)) ds + M^i(t) + A^i(t), \quad t \leq T, \quad (5.22)$$

where M^i is a continuous martingale and A^i is a continuous, increasing process on $(C_S, \mathcal{F}, (\mathcal{F}_t), Q_i)$, with $M^i(0) = A^i(0) = 0$. By Proposition 16.32 in Bass [1], we also have that for $T \geq 0$,

$$\mathbb{E}^{Q_i} [||M^i(T)||^2] < \infty, \quad i = 1, 2.$$

Let W as in (5.17), that is, $W(t) = X(t) - z - \mu t$, and for $i = 1, 2$, let $S^i(W)$ be the class of $\mathbb{R}^2$ -valued processes on $(C_S, \mathcal{F}, (\mathcal{F}_t))$ such that $U \in S^i(X)$ if it has the form

$$U(t) = \int_0^t G(s) dW(s), \quad t \in [0, T],$$

for some 2-dimensional process G such that

$$\mathbb{E}^{\mathbb{Q}_i} \left[\int_0^T \|G(s)\|^2 ds \right] < \infty.$$

Then, by Theorem IV.36 and Corollary 1 to Theorem IV.37 in [36], there exists a $\mathbb{R}^2$ -valued process H^i such that

$$M^i(t) = \int_0^t H^i(s) dW(s) + N^i(t), \quad t \in [0, T],$$

where

- (i) N^i is a square-integrable martingale under $\mathbb{Q}_i$,
- (ii) N^i is strongly orthogonal to every member of $S^i(W)$ under $\mathbb{Q}_i$, that is, $N^i U$ is a $\mathbb{Q}_i$ -martingale for each $U \in S^i(W)$,
- (iii) $\mathbb{E}^{\mathbb{Q}_i} \left[\int_0^T \|H^i(s)\|^2 ds \right] < \infty$.

Now, by (5.22) we have for $t \in [0, T]$,

$$\begin{aligned} f(Z(t)) &= f(z) + \int_0^t \nabla f(Z(s)) \cdot \mu ds + \frac{1}{2} \int_0^t \Delta f(Z(s)) ds \\ &\quad + \int_0^t H^i(s) dW(s) + N^i(t) + A^i(t), \quad \mathbb{Q}_i - \text{a.s.}, \end{aligned} \quad (5.23)$$

hence, by (5.23) we have for $t \in [0, T]$,

$$\begin{aligned} f(Z(t \wedge \tau_n^\delta)) &= f(Z(t \wedge \sigma_n^\delta)) + \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \nabla f(Z(s)) \cdot \mu ds + \frac{1}{2} \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \Delta f(Z(s)) ds \\ &\quad + \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} H^i(s) dW(s) + N^i(t \wedge \tau_n^\delta) - N^i(t \wedge \sigma_n^\delta) + A^i(t \wedge \tau_n^\delta) - A^i(t \wedge \sigma_n^\delta), \end{aligned} \quad (5.24)$$

Q_i -a.s. Now, for each $i = 1, 2$, we have two Doob-Meyer decompositions of the submartingale

$$\{f(Z(t \wedge \tau_n^\delta)) - f(Z(t \wedge \sigma_n^\delta)) - \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \nabla f(Z(s)) \cdot \mu ds - \frac{1}{2} \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \Delta f(Z(s)) ds, t \in [0, T]\},$$

(5.21) and (5.24). Hence, by the uniqueness of the Doob-Meyer decomposition, for each $t \in [0, T]$

$$N^i(t \wedge \tau_n^\delta) - N^i(t \wedge \sigma_n^\delta) + \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} H^i(s) dW(s) = \int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \nabla f(Z(s)) dW(s), \quad (5.25)$$

Q_i - a.s. N_i is strongly orthogonal to every member of $S^i(W)$, and from [36], Theorem IV.37 follows that $\{N^i(t \wedge \tau_n^\delta) - N^i(t \wedge \sigma_n^\delta), t \in [0, T]\}$ is also strongly orthogonal to every member of $S^i(W)$ under Q_i . However, by the above relation, it is also a member of $S^i(W)$, hence it follows that $N^i(t \wedge \tau_n^\delta) - N^i(t \wedge \sigma_n^\delta) = 0$ for $t \in [0, T]$. Then by (5.25) we also have

$$\mathbb{E}^{Q_i} \left[\int_{t \wedge \sigma_n^\delta}^{t \wedge \tau_n^\delta} \|H^i(s) - \nabla f(Z(s))\|^2 ds \right] = 0.$$

Hence,

$$\begin{aligned} & \mathbb{E}^{Q_i} \left[\int_0^t \|H^i(s) - \nabla f(Z(s))\|^2 ds \right] \\ &= \mathbb{E}^{Q_i} \left[\int_0^t \sum_{n=2}^{\infty} \mathbb{1}_{\{s \in [\tau_{n-1}^\delta, \sigma_n^\delta]\}} \|H^i(s) - \nabla f(Z(s))\|^2 ds \right] \\ &\leq \mathbb{E}^{Q_i} \left[\int_0^t \mathbb{1}_{\{Z(s) \in S_{2\delta}^c\}} \|H^i(s) - \nabla f(Z(s))\|^2 ds \right], \end{aligned} \quad (5.26)$$

where $S_{2\delta}^c = S \setminus S_{2\delta}$. Moreover, by the dominated convergence theorem,

$$(5.26) \rightarrow \mathbb{E}^{Q_i} \left[\int_0^t \mathbb{1}_{\{Z(s) \in \partial S\}} \|H^i(s) - \nabla f(Z(s))\|^2 ds \right] = 0 \text{ as } \delta \rightarrow 0,$$

where the last identity is by Lemma 5.3.3 . By (5.23), now follows that for $t \in [0, T]$,

$$\begin{aligned} f(Z(t)) &= f(z) + \int_0^t \nabla f(Z(s)) \cdot \mu ds + \frac{1}{2} \int_0^t \Delta f(Z(s)) ds \\ &+ \int_0^t \nabla f(Z(s)) dW(s) + N^i(t) + A^i(t), \quad Q_i\text{-a.s.} \end{aligned} \quad (5.27)$$

Next, for each $t \geq 0$ let

$$\tilde{\zeta}(t) = \exp \left\{ -\mu \cdot (X(t) - z) + \frac{1}{2} \|\mu\|^2 t \right\},$$

and for each $i = 1, 2$, and $T \geq 0$ define the measure $\tilde{Q}_i^T$ by setting

$$\frac{d\tilde{Q}_i^T}{dQ_i} = \tilde{\zeta}(T). \quad (5.28)$$

Then, under $\tilde{Q}_i^T$, $\{X(t), t \in [0, T]\}$ is a Brownian motion (without drift) started at z , and by (5.27) we have

$$\begin{aligned} f(Z(t)) &= f(z) + \int_0^t \nabla f(Z(s)) dX(s) \\ &+ \frac{1}{2} \int_0^t \Delta f(Z(s)) ds + N^i(t) + A^i(t), \quad Q_i\text{-a.s.} \end{aligned} \quad (5.29)$$

However, note that N^i is also a $\tilde{Q}_i^T$ -martingale on $[0, T]$. Indeed, by (5.28), N^i is a $\tilde{Q}_i^T$ -martingale if $N^i \tilde{\zeta}$ is a Q_i -martingale on $[0, T]$. But this follows since from the fact that N_i is strongly orthogonal to every member of $S^i(X)$ under Q_i , and by its definition $\tilde{\zeta} - 1 \in S^i(W)$ under Q_i . Just like in the proof of Proposition 5.3.2, there exists a probability measure $\tilde{Q}_i$ on $\mathcal{M}$ such that $\tilde{Q}_i(A) = \tilde{Q}_i^T(A)$, whenever $A \in \mathcal{M}_T$. Thus, by (5.29) both $\tilde{Q}_1$ and $\tilde{Q}_2$ satisfy property 2 of Definition 6 with $\mathbb{P}_\mu^z$ replaced by either $\tilde{Q}_1$ or $\tilde{Q}_2$, and μ replaced by the zero vector. $\tilde{Q}_i^T$ and Q_i are mutually absolutely continuous because of (5.28) and because $\tilde{\zeta}(T) > 0$, a.s. under Q_i . Relation (5.20) also follows from (5.28). Properties 1 and 3 of Definition 6 are satisfied if $\mathbb{P}_\mu^z$ is replaced by $\tilde{Q}_i$ because they are satisfied if we replace $\mathbb{P}_\mu^z$ by

Q_i , and we already established that $\tilde{Q}_i^T$ and Q_i are mutually absolutely continuous. Property 1 follows immediately from this. Property 3 can be shown by first showing it with the ∞ in the upper limit of the integral replaced by T , then taking $T \rightarrow \infty$.

□

Proof of Theorem 5.2.1 . In light of Propositions 5.3.2 the only missing part is the proof of uniqueness. Let $z \in S$ and suppose that $\mathbb{P}_1^z$ and $\mathbb{P}_2^z$ are two probability measures satisfying conditions 1, 2, and 3 in Definition 6 of the submartingale problem with drift. Let

$$\mathbb{Q}_1 = \frac{1}{3}\mathbb{P}_1^z + \frac{2}{3}\mathbb{P}_2^z \quad \text{and} \quad \mathbb{Q}_2 = \frac{2}{3}\mathbb{P}_1^z + \frac{1}{3}\mathbb{P}_2^z.$$

Then, one can check that each $\mathbb{Q}_i$ also satisfies conditions 1,2, and 3 in Definition 6 of the submartingale problem with drift. In addition, $\mathbb{Q}_1$ and $\mathbb{Q}_2$ are mutually absolutely continuous. In order to complete the proof, it is therefore sufficient to show that $\mathbb{Q}_1 \equiv \mathbb{Q}_2$. By Lemma 5.3.7 there exist probability measures $\tilde{Q}_i$, $i = 1, 2$, such that properties 1,2, and 3 of Definition 6 are satisfied with P_μ^z replaced by $\tilde{Q}_i$, and μ replaced by the zero vector. The uniqueness result in Section 3.1 of [51] implies $\tilde{Q}_1 = \tilde{Q}_2$. Using the probability measures $\tilde{Q}_i^T$ from the same proposition, we have now that

$$\tilde{Q}_1^T|_{\mathcal{M}_T} = \tilde{Q}_2^T|_{\mathcal{M}_T}.$$

From (5.20) follows that

$$Q_1|_{\mathcal{M}_T} = Q_2|_{\mathcal{M}_T}.$$

Since T was arbitrary, $Q_1 = Q_2$ follows.

□

Proof of Theorem 5.2.2. Recall from the proof of Proposition 5.3.2 that for every $T \geq 0$ there exists the probability measure $\mathbb{P}_{\mu,T}^z$ which is mutually absolutely continuous with respect to $\mathbb{P}_0^z$, and coincides with $\mathbb{P}_\mu^z$ on $\mathcal{M}_T$. By Theorem 2.6 in [31], formulas (5.2) and (5.3) hold for

$\mu = 0$. Then by the mutual absolute continuity of $\mathbb{P}_0^z$ and $\mathbb{P}_{\mu,T}^z$, (5.2) and (5.3) also hold with $\mathbb{P}_\mu^z$ replaced by $\mathbb{P}_{\mu,T}^z$. Since $\mathbb{P}_\mu^z$ and $\mathbb{P}_{\mu,T}^z$ coincide on $\mathcal{M}_T$, both formulas follow.

Proof of Theorem 5.2.3. This follows from Theorem 2.4 in [31] using the measure $\mathbb{P}_{\mu,T}^z$ for every $T > 0$, just like in the proof of Theorem 5.2.2. $\square$

$\square$

Proof of Theorem 5.2.4. Suppose that $1 \leq \alpha < 2$, and Z is a continuous semimartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ for some $z \in S$. Then there exists a decomposition

$$Z(t) = z + M(t) + A(t), \quad t \in [0, \infty), \quad (5.30)$$

where M is a continuous local martingale and A is a finite variation (FV) process on $(C_S, \mathcal{M}, \mathcal{M}_t, \mathbb{P}_\mu^z)$ (see [36], Corollary to Theorem II.31). We know from the proof of Proposition 5.3.2 that for every $T \in [0, \infty)$ there exists a probability measure $\mathbb{P}_{\mu,T}^z$ on $\mathcal{M}$ which is mutually absolutely continuous with respect to $\mathbb{P}_0^z$, and $\mathbb{P}_{\mu,T}^z(A) = \mathbb{P}_\mu^z(A)$ for all $A \in \mathcal{M}_T$. In addition,

$$\frac{d\mathbb{P}_0^z}{d\mathbb{P}_{\mu,T}^z} = \frac{1}{\zeta(T)}$$

where ζ is defined under (5.5). We cast (5.30) in the form

$$Z(t) = z + \tilde{M}(t) + \tilde{A}(t), \quad t \in [0, \infty), \quad (5.31)$$

where

$$\begin{aligned} \tilde{M}(t) &= M(t) - \int_0^t \zeta(s) d \left[\frac{1}{\zeta}, M \right]_s \\ \tilde{A}(t) &= A(t) + \int_0^t \zeta(s) d \left[\frac{1}{\zeta}, M \right]_s. \end{aligned}$$

By the Girsanov-Meyer theorem ([36], Theorem III.31) $\tilde{M}$ is a local martingale on $[0, T]$ and $\tilde{A}$ is a FV process under $\mathbb{P}_0^z$. But this implies that $\tilde{M}$ is a local martingale on $[0, \infty)$ under

$\mathbb{P}_0^z$, hence Z must be a semimartingale under $\mathbb{P}_0^z$, which is in contradiction with the result of [54], Theorem 5.

□

Proof of Theorem 5.2.5. Suppose that $\alpha \geq 2$, let $z \in S$ and suppose that $\mathbb{P}_\mu^z$ is a probability measure on $\mathcal{M}$ satisfying properties 1, 2 and 3 in Definition 6 of the submartingale problem with drift. Selecting $Q_1 = Q_2 = \mathbb{P}_\mu^z$ in Lemma 5.3.7, it follows that there exists a probability measure $\tilde{Q}$ on $\mathcal{M}$ such that properties 1, 2, and 3 of Definition 6 are satisfied with $\mathbb{P}_\mu^z$ replaced by $\tilde{Q}$, and μ replaced by the zero vector. However, this is in direct contradiction with Theorem 3.11 in [51].

□

5.4 Proof of Theorems 5.2.8 and 5.2.6

First, we shall prove Theorem 5.2.8.

Proposition 5.4.1. *The family of probability measures $\{\mathbb{P}_\mu^{z_n}\}$ is tight for any sequence $\{z_n, n \geq 1\}$ in S which converges to some $z \in S$.*

Proof. By Theorem 2.4.10 in [28], it is sufficient to show that

$$\lim_{\delta \downarrow 0} \sup_n \mathbb{P}_\mu^{z_n}(\omega : m^T(\omega, \delta) \geq \varepsilon) = 0, \text{ for any } T > 0, \varepsilon > 0. \quad (5.32)$$

In the above,

$$m^T(\omega, \delta) = \sup_{\substack{|t-s| \leq \delta \\ 0 \leq s, t \leq T}} |\omega(s) - \omega(t)|.$$

Using (5.5) and the Cauchy-Schwarz inequality,

$$\begin{aligned}
& \mathbb{P}_\mu^{z_n}(\omega : m^T(\omega, \delta) \geq \varepsilon) \\
&= \mathbb{E}_0^{z_n}[\mathbb{1}_{\{m^T(\omega, \delta) > \varepsilon\}} \exp\{\mu \cdot (X(T) - X(0)) - \frac{1}{2}||\mu||^2 T\}] \\
&\leq (\mathbb{E}_0^{z_n}[\exp\{2\mu \cdot (X(T) - X(0)) - ||\mu||^2 T\}]\mathbb{P}_0^{z_n}(m^T(\omega, \delta) > \varepsilon))^{1/2} \\
&= \exp\left\{\frac{1}{2}||\mu||^2 T\right\} (\mathbb{P}_0^{z_n}(m^T(\omega, \delta) > \varepsilon))^{1/2}.
\end{aligned} \tag{5.33}$$

By Theorem 3.13 in [51], $\{\mathbb{P}_0^{z_n}\}$ is tight hence

$$\limsup_{\delta \downarrow 0} \limsup_n \mathbb{P}_0^{z_n}(\omega : m^T(\omega, \delta) \geq \varepsilon) = 0, \text{ for any } T > 0, \varepsilon > 0,$$

combining with inequality (5.33), we have (5.32). $\square$

Proof of Theorem 5.2.8. Given Proposition 5.4.1, it only remains to show that any weak limit point $\mathbb{P}_\mu^*$ of the family $\{\mathbb{P}_\mu^{z_n}\}$ is a solution to the submartingale problem starting from z , then by the uniqueness part of Theorem 5.2.1, $\mathbb{P}_\mu^{z_n} \Rightarrow \mathbb{P}_\mu^z$ as $n \rightarrow \infty$.

It is straightforward that $\mathbb{P}_\mu^*$ satisfies condition 1 in Definition 6 (the submartingale problem), since for any $k \geq 1$ and the closed set $C_k = \{\omega \in C_S : |\omega(0) - z| \leq \frac{1}{k}\}$, $1 = \limsup_n \mathbb{P}_\mu^{z_n}(C_k) \leq \mathbb{P}_\mu^*(C_k)$ hence $\mathbb{P}_\mu^*$ concentrates on $\{\omega \in C_S : \omega(0) = z\}$. The condition 2 is also satisfied since the submartingale property is preserved under the weak convergence. Now we prove $\mathbb{P}_\mu^*$ satisfies condition 3, we need to show that if $(z_n, n \geq 1) \subset S$, $z \in S$ such that $\lim_{n \rightarrow \infty} z_n = z$ and $\mathbb{P}_\mu^{z_n} \Rightarrow \mathbb{P}_\mu^*$, then

$$\mathbb{E}_\mu^* \left[\int_0^\infty \mathbb{1}_{\{(Z(t))=0\}} dt \right] = 0. \tag{5.34}$$

Let $\epsilon > 0$ and $t \geq 0$ be arbitrary. Under the local uniform topology the event $\{w \in C_S : |Z_t(w)| < \epsilon\}$ is an open set, and $\{w \in C_S : |Z_t(w)| \leq \epsilon\}$ is a closed set. By the Portmanteau

Theorem, (5.5), and the Cauchy-Schwarz inequality

$$\begin{aligned} \mathbb{P}_\mu^* (|Z_t| < \epsilon) &\leq \liminf_n \mathbb{P}_\mu^{z_n} (|Z_t| < \epsilon) = \liminf_n \mathbb{E}_0^{z_n} [\zeta(t) 1_{\{|Z_t| < \epsilon\}}] \leq \\ \liminf_n (\mathbb{E}_0^{z_n} [\zeta(t)^2] \mathbb{P}_0^{z_n} (|Z_t| < \epsilon))^{1/2} &\leq \exp \left\{ \frac{1}{2} |\mu|^2 t \right\} \limsup_n (\mathbb{P}_0^{z_n} (|Z_t| \leq \epsilon))^{1/2} \leq \\ \exp \left\{ \frac{1}{2} |\mu|^2 t \right\} (\mathbb{P}_0^z (|Z_t| \leq \epsilon))^{1/2}. \end{aligned}$$

The last inequality follows from the Feller property for the drift-less case ([51], Theorem 3.13). We now let $\epsilon \downarrow 0$, and then from property 3 in the definition of the Submartingale Problem follows that

$$\mathbb{P}_\mu^* (Z_t = 0) = 0,$$

for Lebesgue-almost every $t \geq 0$. Identity (5.34) follows. □

Proof of Theorem 5.2.6. This follows in the standard way from the uniqueness of the solution to the submartingale problem, using the regular conditional probability measures for $\mathcal{M}$ given $\mathcal{M}_\tau$ under $\mathbb{P}_\mu^z$. In particular, Lemma 3.1, and Corollary 3.3 in [51] remain true in the presence of a drift, without changing a single word in their proofs. Then the strong Markov property follows, again exactly the same way as in [51], Theorem 3.14. □

5.5 Proof of Theorem 5.2.9

Proof of Theorem 5.2.9. It is sufficient to show that the $\hat{C}(S)$ -Feller property holds in the case when the drift is zero. Indeed, suppose that for the driftless case the $\hat{C}(S)$ -Feller property holds. By Theorem 1.10 in Bottcher, Schilling, and Wang [2], the $\hat{C}(S)$ -Feller property holds for $\{\mathbb{P}_\mu^z, \forall z \in S\}$ if and only if there exists an increasing sequence of bounded sets $B_n \in \mathcal{B}(S)$

with $\cup_{n \geq 1} B_n = S$ such that for every $t > 0$ and $n \geq 1$

$$\lim_{|z| \rightarrow \infty} \mathbb{P}_\mu^z(Z(t) \in B_n) = 0.$$

Since we already know that uniqueness holds for the submartingale problem with drift μ , we may assume that $\{\mathbb{P}_\mu^z, z \in S\}$ is exactly the family we created in the existence part of this paper using Girsanov's theorem. By Theorem 1.10 in [2], there exists an increasing sequence of bounded sets $B_n \in \mathcal{B}(S)$ with $\cup_{n \geq 1} B_n = S$ such that for every $t > 0$ and $n \geq 1$,

$$\lim_{|z| \rightarrow \infty} \mathbb{P}_0^z(Z(t) \in B_n) = 0,$$

where $\{\mathbb{P}^z, z \in S\}$ is the solution of the submartingale problem without drift. By formula (5.5) we have that

$$\begin{aligned} \mathbb{P}_\mu^z(Z(t) \in B_n) &= \mathbb{E}_0^z[\zeta(t) \mathbf{1}_{\{Z(t) \in B_n\}}] \leq (\mathbb{E}_0^z[(\zeta(t))^2] \mathbb{P}^z(Z(t) \in B_n))^{1/2} \\ &= \exp\left\{\frac{\mu^2}{2}t\right\} (\mathbb{P}_0^z(Z(t) \in B_n))^{1/2}, \end{aligned}$$

which shows that the $\hat{C}(S)$ -Feller property holds for $\{\mathbb{P}_\mu^z, \forall z \in S\}$.

In the rest of this proof, we shall show that in the case of $\mu = 0$ the $\hat{C}(S)$ -Feller property holds. Let X be the process identified in Proposition 5.3.1. It has been shown in Williams and Varadhan [51] that the Feller property holds, hence the $C_b(S)$ -Feller property holds, so by Theorem 1.10 in [2], it is sufficient to show that for every $t > 0$

$$\mathbb{P}_0^z(Z(t) \in B_n) \rightarrow 0$$

as $|z| \rightarrow \infty$, where $B_n = \{z \in S : |z| \leq n\}$, for $n \geq 1$. Note that

$$\mathbb{P}_0^z(Z(t) \in B_n) = \mathbb{P}_0^z(Z(t) \in B_n, \tau \leq t) + \mathbb{P}_0^z(Z(t) \in B_n, \tau > t),$$

where

$$\tau = \inf\{t \geq 0 : Z(t) \in \partial S\}.$$

We treat the second term first. It is bounded above by

$$\mathbb{P}_0^z(X(t) \in B_n) \leq \mathbb{P}_0^z(|X(t) - z| \geq |z| - n) = \Pr(|w(t)| > |z| - n) \rightarrow 0,$$

as $|z| \rightarrow \infty$, where w is a standard 2-dimensional Brownian motion started at zero. Next, we treat the first term. Let $T_n = \inf\{t \geq 0 : Z(t) \in B_n\}$. Then the first term can be written as a sum of three terms

$$\begin{aligned} & \mathbb{P}_0^z \left(Z(t) \in B_n, \tau \leq t, |Z(\tau)| < \frac{|z|}{2} \right) + \\ & \mathbb{P}_0^z \left(Z(t) \in B_n, \tau \leq t, T_n \leq \tau, |Z(\tau)| \geq \frac{|z|}{2} \right) + \\ & \mathbb{P}_0^z \left(Z(t) \in B_n, \tau \leq t, T_n > \tau, |Z(\tau)| \geq \frac{|z|}{2} \right) \end{aligned}$$

Again, we treat the three terms separately. By Proposition 5.3.1 we have that $Z(\tau) = X(\tau)$, where X is a standard Brownian motion with zero drift started at z under $\mathbb{P}_0^z$, thus first term is bounded above by

$$\mathbb{P}_0^z \left(\tau \leq t, |X(\tau)| < \frac{|z|}{2} \right) \leq \mathbb{P}_0^z \left(\tau \leq t, |X(\tau) - z| > \frac{|z|}{2} \right) \leq \Pr \left(\max_{s \leq t} |w(s)| > \frac{|z|}{2} \right) \rightarrow 0,$$

as $|z| \rightarrow \infty$. The second term is bounded above by

$$\mathbb{P}_0^z(T_n \leq \tau \leq t) \leq \mathbb{P}_0^z(X - z \text{ reaches } B_n - z \text{ by time } t) = \Pr(w \text{ reaches } B_n - z \text{ by time } t) \rightarrow 0$$

as $|z| \rightarrow \infty$. Here w is a standard 2-dimensional Brownian motion started at the origin.

For analyzing the third term we define the stopping time $T_n^\tau = \inf\{t \geq \tau : Z(t) \in B_n\}$.

The third term is bounded above by

$$\begin{aligned} \mathbb{P}_0^z \left(\tau < T_n \leq t, |Z(\tau)| > \frac{|z|}{2} \right) &\leq \mathbb{P}_0^z \left(\tau < T_n \leq \tau + t, |Z(\tau)| > \frac{|z|}{2} \right) \leq \\ &\mathbb{P}_0^z \left(T_n^\tau \leq t, |Z(\tau)| > \frac{|z|}{2} \right). \end{aligned}$$

By the strong Markov property, this can be written as

$$\int_{\partial S \cap B_{|z|/2}^c} \mathbb{P}_0^x(T_n \leq t) P_0^z(Z(\tau) \in dx).$$

By the scaling property (Lemma 2.1 in [55]), the process $\{Z(t), t \geq 0\}$ under $\mathbb{P}_0^x$ induces the same measure on $\mathcal{M}$ as $\{|x|Z(t/|x|^2), t \geq 0\}$ induces under $\mathbb{P}_0^{x/|x|}$, for every non-zero $x \in S$.

Then the above expression can be written as

$$\begin{aligned} &\int_{\partial S \cap B_{|z|/2}^c} \mathbb{P}_0^{x/|x|} (|x|^2 T_{n/|x|} \leq t) \mathbb{P}_0^z(Z(\tau) \in dx) = \\ &\int_{\partial S_1 \cap B_{|z|/2}^c} \mathbb{P}_0^{u_1} (|x|^2 T_{n/|x|} \leq t) \mathbb{P}_0^z(Z(\tau) \in dx) + \\ &\int_{\partial S_2 \cap B_{|z|/2}^c} \mathbb{P}_0^{u_2} (|x|^2 T_{n/|x|} \leq t) \mathbb{P}_0^z(Z(\tau) \in dx), \end{aligned}$$

where u_1 and u_2 are the unit vectors $u_1 = (1, 0)$, and $u_2 = (\cos \xi, \sin \xi)$. By symmetry, it is sufficient to show that the first term converges to 0 as $|z| \rightarrow \infty$. If $|z|/2 > 2n$, then it is bounded above by

$$\sup_{|x| > |z|/2} \mathbb{P}_0^{u_1} (|x|^2 T_{1/2} \leq t) = \mathbb{P}_0^{u_1} \left(\frac{|z|^2}{4} T_{1/2} \leq t \right) \rightarrow 0,$$

as $|z| \rightarrow \infty$, which completes the proof of the proposition.

□

5.6 Proof of Theorem 4.1.1

Before proving the existence part of Theorem 4.1.1, we must first establish some preliminary results. Let $\{B_t, t \geq 0\}$ be the coordinate mapping process on $C(\mathbb{R}_+, \mathbb{R}^2)$, whose natural filtration is given by $\mathcal{W}_t = \sigma(B_s, 0 \leq s \leq t)$ for $t \geq 0$, and let $\mathcal{W} = \sigma(B_s, s \geq 0)$. Recall that v_i is the reflection direction on ∂S_i for $i = 1, 2$, and let R be the 2×2 matrix defined by $R_{ij} =$ the i -th component of v_j . The following result is adapted from Theorem 3.1 in [53].

Proposition 5.6.1. *For any $w \in C(\mathbb{R}_+, \mathbb{R}^2)$ with $w(0) \in S$, there exists a unique triple (ϕ, η, T_0) , where $\phi \in C_S$, $\eta \in C(\mathbb{R}_+, [0, +\infty]^2)$ and $T_0 : C(\mathbb{R}_+, \mathbb{R}^2) \rightarrow [0, +\infty]$, satisfying the following four conditions,*

1. $\phi(t) = w(t) + R\eta(t)$ for each $t \in [0, T_0)$;
2. $\phi(t) \neq 0$ for all $t < T_0$ and $\phi(t) = 0$ for all $t \geq T_0$;
3. For $j = 1, 2$, $\eta(0) = 0$ and $\eta_j(\cdot)$ is non-decreasing and finite for $t \in [0, T_0)$;
4. For $j = 1, 2$, η_j only increases when $\phi(t)$ is on $\partial S_j \setminus \{0\}$.

Furthermore, we have the following two properties

- (i) T_0 is a stopping time on $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}, \mathcal{W}_t)$;
- (ii) Define the map $\Gamma : C(\mathbb{R}_+, \mathbb{R}^2) \mapsto C_S$ such that $\Gamma(w) = \phi$. Then, $\tau_0 \circ \Gamma = T_0$, the map $\Gamma_t \equiv \Gamma(\cdot)(t)$ is $\mathcal{W}_t$ -measurable and Γ is $\mathcal{W}/\mathcal{M}$ -measurable.

Now for each $z \in S$, let $\hat{\mathbb{P}}_0^z$ be the unique measure on $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}, \mathcal{W}_t)$ under which $\{B_t, t \geq 0\}$ is a standard Brownian motion started at z (the subscript 0 indicates that the Brownian motion has zero drift under $\hat{\mathbb{P}}_0^z$). Next, for each $\mu \in \mathbb{R}^2$ and $T \geq 0$, define the measure $\hat{\mathbb{P}}_{\mu, T}^z$ on $\mathcal{W}_T$ by

$$\frac{d\hat{\mathbb{P}}_{\mu, T}^z}{d\hat{\mathbb{P}}_0^z} = \exp \left(\mu(B_T - z) - \frac{1}{2} \|\mu\|^2 T \right),$$

and define also

$$\hat{B}_t = B_t - \mu t,$$

and note that $\hat{B}$ is a standard 2 dimensional Brownian motion started at z under $\hat{\mathbb{P}}_{\mu,T}^z$ on $[0, T]$. Then, by Theorem 4.2 in [34] there exists a measure $\hat{\mathbb{P}}_\mu^z$ on $\mathcal{W}$ which coincides with $\hat{\mathbb{P}}_{\mu,T}^z$ on $\mathcal{W}_T$ for all $T \geq 0$. For every $z \in S$ and $\mu \in \mathbb{R}^2$ the process B is a Brownian motion with drift μ started at z under the probability measure $\hat{\mathbb{P}}_\mu^z$. Moreover, since by Proposition 5.6.1, Γ is a measurable map from $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}, \mathcal{W}_t)$ to $(C_S, \mathcal{M}, \mathcal{M}_t)$, we may denote by $\bar{\mathbb{P}}_\mu^z$ the measure induced on $\mathcal{M}$ by the mapping Γ under $\hat{\mathbb{P}}_\mu^z$, i.e.,

$$\bar{\mathbb{P}}_\mu^z(A) \equiv \hat{\mathbb{P}}_\mu^z(\Gamma^{-1}(A)), \text{ for each } A \in \mathcal{M}. \quad (5.35)$$

We next prove that the family of measures $\{\bar{\mathbb{P}}_\mu^z, z \in S\}$ is a solution to the absorbed process problem of Definition 5. Since conditions 1 and 3 of Definition 5 are trivially satisfied by $\{\bar{\mathbb{P}}_\mu^z, z \in S\}$, it remains to prove that condition 2 is satisfied as well. This is achieved in Lemma 5.6.2.

Lemma 5.6.2. *Let the family of measures $\{\bar{\mathbb{P}}_\mu^z, z \in S\}$ be defined as in (5.35) and let Z be the coordinate-mapping process on $(C_S, \mathcal{M}, \mathcal{M}_t)$. Then, the process*

$$\left\{ f(Z(t \wedge \tau_0)) - \int_0^{t \wedge \tau_0} \mu \cdot \nabla f(Z(s)) ds - \frac{1}{2} \int_0^{t \wedge \tau_0} \Delta f(Z(s)) ds, t \geq 0 \right\} \quad (5.36)$$

is a submartingale on $(C_S, \mathcal{M}, \mathcal{M}_t, \bar{\mathbb{P}}_\mu^z)$, for each $f \in C_b^2(S)$ such that $D_i f \geq 0$ on ∂S_i for $i = 1, 2$.

Proof. For each $w \in C(\mathbb{R}_+, \mathbb{R}^2)$, let $\phi(w) = \Gamma(w)$ and note that by Proposition 5.6.1 we may write $\phi(t) = w(t) + R\eta(t)$ for all $t \geq 0$. Now, on $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}_t, \mathcal{W})$, we consider the process $\{\phi(t), t \geq 0\}$ with $\phi(t) = \Gamma_t(w)$ for any $w \in C(\mathbb{R}_+, \mathbb{R}^2)$. Recall that the coordinate mapping process $\{B_t, t \geq 0\}$ is a Brownian motion on $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}_t, \mathcal{W})$ under $\hat{\mathbb{P}}^z$, and $\{B_t - \mu t, t \geq 0\}$ is a Brownian motion on $(C(\mathbb{R}_+, \mathbb{R}^2), \mathcal{W}_t, \mathcal{W})$ under $\hat{\mathbb{P}}_\mu^z$. Notice, by Theorem

1 in [54], that $R\eta(t)$ is of finite variation on $[0, t \wedge T_0]$ for any $t \geq 0$, and $w(t) = B(w)(t)$, we get that $\{\phi(t), t \geq 0\}$ is a semimartingale under $\hat{\mathbb{P}}^z$. On the other hand, the Girsanov transform keeps the semimartingale property, so $\{\phi(t), t \geq 0\}$ is also a semimartingale under $\hat{\mathbb{P}}_\mu^z$. Hence, for each $f \in C_b^2(S)$ such that $D_i f \geq 0$ on ∂S_i for $i = 1, 2$, we use Itô's formula under $\hat{\mathbb{P}}_\mu^z$ and get

$$\begin{aligned} f(\phi(t \wedge T_0)) - f(\phi(0)) &= \sum_{i=1}^2 \int_0^{t \wedge T_0} \frac{\partial f}{\partial x_i}(\phi(s)) d(w_i(s) - \mu_i s) + \int_0^{t \wedge T_0} \mu \cdot \nabla f(\phi(s)) ds \\ &+ \int_0^{t \wedge T_0} (D_1 f(\phi(s)), D_2 f(\phi(s))) \cdot d\eta(s) + \frac{1}{2} \int_0^{t \wedge T_0} \Delta f(\phi(s)) ds \end{aligned}$$

Since we have by Proposition 5.6.1 that for $i = 1, 2$,

$$d\eta_i(s) = \mathbf{1}_{\{\phi(s) \in \partial S_i \setminus \{0\}\}} d\eta(s), \quad s \geq 0,$$

and by the assumption on f that for $i = 1, 2$,

$$D_i f(\phi(s)) \mathbf{1}_{\{\phi(s) \in \partial S_i \setminus \{0\}\}} \geq 0, \quad s \geq 0,$$

it follows that the process

$$\left\{ \int_0^{t \wedge T_0} (D_1 f(\phi(s)), D_2 f(\phi(s))) \cdot d\eta(s), t \geq 0 \right\}$$

is increasing. On the other hand, since $\{B_t - \mu t, t \geq 0\}$ is a Brownian motion under $\hat{\mathbb{P}}_\mu^z$, the process

$$\left\{ \sum_{i=1}^2 \int_0^{t \wedge T_0} \frac{\partial f}{\partial x_i}(\phi(s)) d(w_i(s) - \mu_i s), t \geq 0 \right\}$$

is a martingale under $\hat{\mathbb{P}}_\mu^z$, so

$$\begin{aligned} & f(\phi(t \wedge T_0)) - \int_0^{t \wedge T_0} \mu \cdot \nabla f(\phi(s)) ds - \frac{1}{2} \int_0^{t \wedge T_0} \Delta f(\phi(s)) ds \\ &= f(\phi(0)) + \sum_{i=1}^2 \int_0^{t \wedge T_0} \frac{\partial f}{\partial x_i}(\phi(s)) d(w_i(s) - \mu_i s) \\ &+ \int_0^{t \wedge T_0} (D_1 f(\phi(s)), D_2 f(\phi(s))) \cdot d\eta(s) \end{aligned}$$

is a submartingale under $\hat{\mathbb{P}}_\mu^z$. It follows from (5.35) that the process under (5.36) is also a submartingale under the induced measure $\bar{\mathbb{P}}_\mu^z$. $\square$

Proof of the existence part of Theorem 4.1.1. The existence of a solution to the absorbed process problem follows from Lemma 5.6.2. $\square$

Proof of the uniqueness part of Theorem 4.1.1. The proof of the uniqueness of the solution to the absorbed process problem is very similar to that of the solution of the submartingale problem, hence in this section we shall state the appropriate lemmas and indicate the necessary changes in order to adapt the proofs in Section 5.3 to the absorbed process problem.

Lemma 5.6.3. *Let $\alpha \in \mathbb{R}$ arbitrary, and suppose that $\{\mathbb{P}_\mu^{z,0}, z \in S\}$ is a solution to the absorbed process problem with drift $\mu \in \mathbb{R}^2$. Then, for all $z \in S$,*

$$\mathbb{E}_\mu^{z,0} \left[\int_0^{\tau_0} \mathbf{1}_{\{Z(t) \in \partial S\}} dt \right] = 0.$$

Proof. The proof is almost identical to that of Lemma 5.3.3 with the modification that all processes must be stopped at τ_0 . $\square$

Lemma 5.6.4. *Let $\alpha \in \mathbb{R}$ be arbitrary. Suppose that $\{\mathbb{P}_\mu^{z,0}, z \in S\}$ is a solution to the absorbed process problem with drift $\mu \in \mathbb{R}^2$. Then there exists a process X on $(C_S, \mathcal{M}, \mathcal{M}_t)$ such that for all $z \in S$, X is a Brownian Motion with drift μ under $\mathbb{P}_\mu^{z,0}$ started at z and stopped at τ_0 . In addition, $Y = Z - X$ is flat on $[\sigma_n^\delta, \tau_n^\delta]$.*

Proof. The proof is very similar to the proofs of Lemmas 5.3.4, 5.3.5, and 5.3.6, with the difference that all processes must be stopped at τ_0 . $\square$

Proof of the uniqueness part of Theorem 4.1.1. The proof is basically a copy of the proof of the uniqueness of the solution to the submartingale problem (Lemma 5.3.7 and the proof of Theorem 5.2.1). The necessary changes in order to adapt that proof to the present situation are the following:

- All processes must be stopped at τ_0 ;
- The constraint that the test function f is constant in a neighborhood of the vertex must be erased (above (5.21) in the adapted proof;
- At the end of the proof, instead of using the uniqueness of the solution to the submartingale problem with no drift, we must use the uniqueness of the solution to the absorbed process problem with no drift ([51], Theorem 2.1). $\square$

Appendix A

The construction of a special test function

A.1

Lemma A.1.1. *There exists a function $f_{\varepsilon,C} \in C_b^2(S)$ satisfying*

$$f_{\varepsilon,C}(x, y) = \begin{cases} 0, & \text{if } (x, y) \in S \setminus S^{\varepsilon/3}, \\ y, & \text{if } (x, y) \in S^{2\varepsilon/3}, y \leq C, \end{cases}$$

such that in addition $f_{\varepsilon,C}(x, 0) = 0$ for all $x \geq 0$, and $D_i f_{\varepsilon,C} \geq 0$ on ∂S_i .

Proof. Let $h_1 \in C_b^2(\mathcal{R})$ such that $h_1(x) \geq 0$ for all $x \in \mathbb{R}$ and

$$h_1(x) = \begin{cases} 0, & \text{if } x \leq \varepsilon/3, \\ 1, & \text{if } x \geq 2\varepsilon/3. \end{cases}$$

Let $h_2 \in C_b^2(\mathcal{R})$ such that $h_2(y) = y$ if $y \leq C$. Then

$$f_{\varepsilon,C} = h_1 \left(x - \frac{y}{\tan \xi} \right) h_2(y) \quad (x, y) \in S$$

satisfies the requirements of the lemma. Note that for any $\delta > 0$ we have $(x, y) \in S^\delta$ if and only if $x - y/\tan \xi \geq \delta$. Using this fact repeatedly one can verify the above statement by straightforward calculation. $\square$

Remark A.1.2. There are slightly different definitions for the term “augmented filtration” in the literature; we use this term as defined in [41], Definition II.67.3. Let $\mathbb{P}$ be an arbitrary probability measure on $\mathcal{M}$, and let $(C_S, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be the augmentation of the probability space $(C_S, \mathcal{M}, (\mathcal{M}_t), \mathbb{P})$, in the above sense. It is known that right-continuous martingales (submartingales) on $(C_S, \mathcal{M}, (\mathcal{M}_t), \mathbb{P})$ are also right-continuous martingales (submartingales) on $(C_S, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ (Lemma II.67.10 in [41]). The probability measure $\mathbb{P}$ in the second probability space is the extension of $\mathbb{P}$ from $\mathcal{M}$ to $\mathcal{F}$, without changing the notation. Also, it follows from the martingale characterization of Brownian motion (Theorem 3.3.16 in [28]) that a Brownian motion on $(C_S, \mathcal{M}, (\mathcal{M}_t), \mathbb{P})$ is also a Brownian motion on $(C_S, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

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